Genetics and Genomics
Medicine

Reviewed Preprint
v2 • August 23, 2024
Revised by authors

Reviewed Preprint
v1 • May 28, 2024

Biallelic pathogenic variants in *DNAH3* cause male infertility in humans and mice

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Abstract

Axonemal protein complexes, including the outer and inner dynein arms (ODA/IDA), are highly ordered structures of the sperm flagella that drive sperm motility. Deficiencies in several axonemal proteins have been associated with male infertility, which is characterized by asthenozoospermia or asthenoteratozoospermia. Dynein axonemal heavy chain 3 (DNAH3) resides in the IDA and is highly expressed in the testis. However, the relationship between DNAH3 and male infertility is still unclear. Herein, we identified biallelic variants of *DNAH3* in four unrelated Han Chinese infertile men with asthenoteratozoospermia through whole-exome sequencing (WES). These variants contributed to deficient DNAH3 expression in the patients' sperm flagella. Importantly, the patients represented the anomalous sperm flagellar morphology, and the flagellar ultrastructure was severely disrupted. Intriguingly, *Dnah3* knockout (KO) male mice were also infertile, especially showing the severe reduction in sperm movement with the abnormal IDA and mitochondrion structure. Mechanically, nonfunctional DNAH3 expression resulted in decreased expression of IDA-associated proteins in the spermatozoa flagella of patients and KO mice, including DNAH1, DNAH6, and DNALI1, the deletion of which has been involved in disruption of sperm motility. Moreover, the infertility of patients with *DNAH3* variants and *Dnah3* KO mice could be rescued by intracytoplasmic sperm injection (ICSI) treatment. Our findings indicated that *DNAH3* is a novel pathogenic gene for asthenoteratozoospermia and may further contribute to the diagnosis, genetic counseling, and prognosis of male infertility.

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eLife assessment

This **valuable** study identifies biallelic variants of DNAH3 in unrelated infertile men and reports infertility in DNAH3 knockout mice. The authors demonstrate that compromised DNAH3 activity decreases the expression of IDA-associated proteins in the spermatozoa of human patients and knockout mice, providing **convincing** evidence that DNAH3 is a novel pathogenic gene for asthenoteratozoospermia and male infertility. The study will be of substantial interest to clinicians, reproductive counselors, embryologists, and basic researchers working on infertility and assisted reproductive technology.

https://doi.org/10.7554/eLife.96755.2.sa4

Introduction

Infertility is a global public health and social problem that affects approximately one in six couples worldwide (1). Male infertility, which accounts for half of infertile cases, is a multifactorial disease with common phenotypes, including oligo/azoospermia (poor sperm count or absence of spermatozoa); teratozoospermia (aberrant sperm morphology); asthenozoospermia (weakened sperm motility); and a combination of these phenotypes, such as asthenoteratozoospermia, oligoasthenozoospermia, oligoteratozoospermia and oligoasthenoteratozoospermia (2, 3).

Asthenoteratozoospermia is one of the most common phenotypes of male infertility, and genetic factors have been established as the predominant cause of asthenoteratozoospermia. Multiple morphological abnormalities of the flagella (MMAF), a subtype of asthenoteratozoospermia, characterized by a mosaic of abnormalities of the flagellar morphology, including absent, short, coiled, bent and/or irregular flagella, is almost always caused by genetic defects (4, 5). To date, more than 40 genes have been identified as pathogenic genes of MMAF, but these genes can only explain approximately 60% of MMAF-affected cases (6–9). Therefore, the genetic basis of the remaining cases is still unknown.

The motility of a sperm is driven by its rhythmically beating flagella, and at the center of the flagella lies a conserved axonemal structure, containing the “9 + 2” microtubular arrangement: a ring of nine microtubule doublets (MTDs) surrounding a central pair (CP) of singlet microtubules. Each MTD consists of an A tubule and a B tubule, and the outer (ODA) and inner (IDA) dynein arms are anchored along the A tubule (10). The ODA and IDA are ATPase-based protein complexes that drive the movement between the A tubule and the neighboring B tubule of the next doublet, producing the original force for sperm motility (11, 12). Structural and functional abnormalities of the ODA and IDA have been demonstrated to cause male infertility associated with asthenozoospermia and/or asthenoteratozoospermia (4, 13, 14).

The dynein axonemal heavy chain (DNAH) family comprises a series of proteins (DNAH1–3, DNAH5–12, and DNAH17) that are precisely assembled with other axonemal dynein motor proteins in the ODAs and IDAs of sperm flagella and motile cilia (15–17). In humans, DNAH1, DNAH2, DNAH6, DNAH7, DNAH8, DNAH10, DNAH12 and DNAH17 are highly expressed in the testis, and deficiency of these proteins has been demonstrated to cause MMAF-associated asthenoteratozoospermia (18–25). DNAH3 is an evolutionarily conserved IDA-associated protein and is highly expressed in testes of humans and mice (26). Deficient DNAH3 has been shown to impair sperm motility in *Drosophila* and cattle (27, 28). In humans, *DNAH3* has been identified as a novel breast cancer candidate gene (29). However, the role of DNAH3 in male reproduction in humans and mice remains largely unknown.

In the present study, we identified four biallelic variations in *DNAH3* in four unrelated Han Chinese patients with asthenoteratozoospermia using whole-exome sequencing (WES). The spermatozoa of the patients showed extremely reduced sperm motility and a high proportion of sperm tail defects characterized by the MMAF phenotype. We further generated *Dnah3* knockout (KO) mice, and the male KO mice expectedly showed aberrations in sperm movement, flagellar IDA, and mitochondrion. Moreover, the absence of DNAH3 led to decreased expression of other IDA-associated proteins, including DNAH1, DNAH6 and DNALI1. Importantly, good outcomes of intracytoplasmic sperm injection (ICSI) treatment were observed in *DNAH3*-deficient patients and *Dnah3* KO mice. This study revealed *DNAH3* as a novel pathogenic gene of asthenoteratozoospermia, and the findings provide valuable suggestions for the clinical diagnosis and treatment of male infertility.

Results

Identification of biallelic pathogenic variants of *DNAH3* in four unrelated infertile men

In the present study, we employed whole-exome sequencing (WES) to identify potential candidate variants associated with primary asthenoteratozoospermia. After comprehensive filtering and screening, we identified 98, 101, 67 and 91 candidate variants for Patient 1, Patient 2, Patient 3 and Patient 4, respectively (Table S1). To refine these candidate variants, we excluded those whose corresponding genes were not expressed in the human or mouse testis, were associated with diseases unrelated to male infertility, or were monoallelic variants. Ultimately, only bi-allelic variants in *DNAH3* (NG_052617.1, NM_017539.2, NP_060009.1) remained, suggesting as the pathogenic variants responsible for the infertility of the patients: a compound heterozygous mutation of c.3590C>T (p.Pro1197Leu) and c.3590C>G (p.Pro1197Arg) in Patient 1, a homozygous missense mutation of c.4837G>T (p.Ala1613Ser) in Patient 2, a compound heterozygous mutation of c.5587del (p.Leu1863*) and c.10355C>T (p.Ser3452Leu) in Patient 3 and a compound heterozygous mutation of c.2314C>T (p.Arg772Trp) and c.4045G>A (p.Asp1349Asn) in Patient 4 (**Figure 1A**). Importantly, routine semen analysis revealed that all patients showed extremely reduced sperm motility and a high proportion of sperm tail defects (**Table 1**). These variants either were not recorded or had an extremely low frequency in East Asian population in multiple public population databases, including the ExAC browser, GnomAD and the 1000 Genomes Project, and were predicted to be potentially deleterious by SIFT (*https://sift.bii.a-star.edu.sg/*), PolyPhen-2 (*http://genetics.bwh.harvard.edu/pph2/*), MutationTaster (*https://www.mutationtaster.org/*), and CADD (*https://cadd.gs.washington.edu/*) (**Table 2**) (30–33). Next, Sanger sequencing confirmed these variants in the probands, and their fertile parents carried the heterozygous variants (**Figure 1A**). Moreover, the variant sites are localized in several domains of the DNAH3 protein and are highly conserved across species (**Figure 1B**).

Strikingly, immunofluorescence staining revealed that DNAH3 was exclusively resided in the tail and concentrated in the midpiece of control sperm. However, the fluorescence signal of DNAH3 was hardly detected in the patients' spermatozoa (**Figure 1C**). Additionally, subsequent western blotting analysis yielded consistent results with immunofluorescence staining, indicating that these variants led to disrupted expression of DNAH3 (**Figure 1D**). These results suggested that biallelic variants in *DNAH3* disrupted DNAH3 expression and might be responsible for the infertility of the four patients.

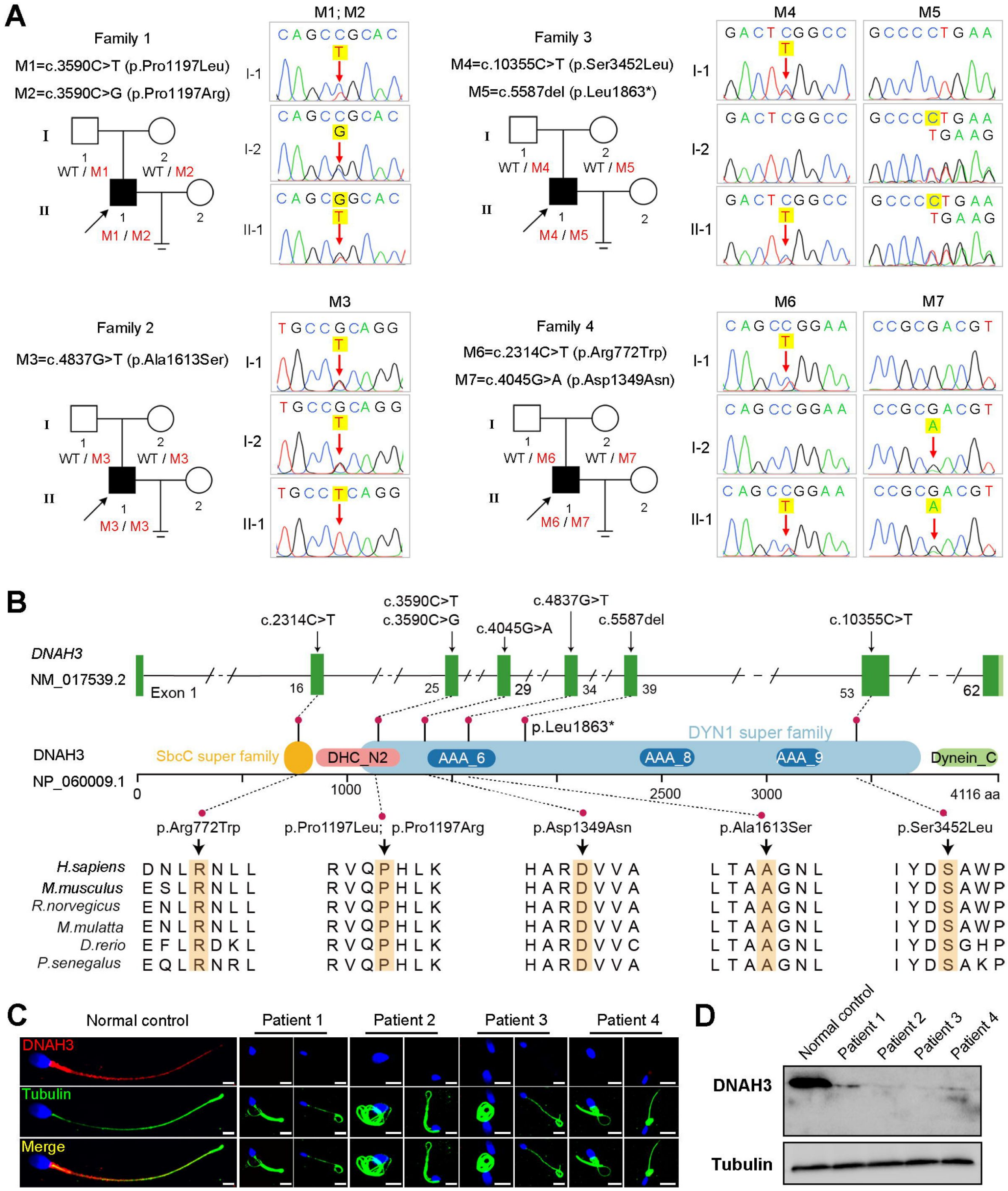


Figure 1.

Identification of biallelic pathogenic variants in *DNAH3* from four unrelated infertile families.

(**A**) Pedigrees of four families affected by *DNAH3* variants (M1–M7). Black arrows indicate the probands in these families. (**B**) Location of the variants and conservation of affected amino acids in DNAH3. Black arrows indicate the position of the variants. (**C**) Immunofluorescence staining of DNAH3 in sperm from the patients and normal control. Red, DNAH3; green, α-Tubulin; blue, DAPI; scale bars, 5 μm. (**D**) Western blotting analysis of DNAH3 expressed in spermatozoa from the patients and normal control.

Table 1.

Semen analysis of the patients in the present study.

		Patient 1	Patient 2	Patient 3	Patient 4	Reference
Semen parameters	Semen volume (ml)	4.3	3.3	0.8	4.6	≥1.5
	Semen concentration (10^6/ml)	2.0	0.5	11.0	21.0	≥15.0
	Motility (%)	6.0	3.0	0	0	≥40.0
	Progressive motility (%)	0	2.3	0	0	≥32.0
Sperm morphology	Normal (%)	1.3	1.1	1.8	1.2	≥4.0
	Tail defects (%)	91.3	87.5	96.5	97.5	-

-, not applicable.

Table 2.

Variants analysis of the patients in the present study.

		Patient 1		Patient 2	Patient 3		Patient 4	
Variant	cDNA mutation [1]	c.3590C>T	c.3590C>G	c.4837G>T	c.5587del	c.10355C>T	c.2314C>T	c.4045G>A
	Protein alteration	p.Pro1197Leu	p.Pro1197Arg	p.Ala1613Ser	p.Leu1863*	p.Ser3452Leu	p.Arg772Trp	p.Asp1349Asn
	Mutation type	Missense	Missense	Missense	Nonsense	Missense	Missense	Missense
Allele frequency	ExAC_EAS	0.0001	0	0.004165	0	0.0006	0.0019	0.0065
	GnomAD_EAS	0.00005016	0	0.00277415	0	0.0008	0.002	0.007
	1000 Genomes Project_EAS	0	0	0.0050	0	0	0.0040	0.0069
Function prediction	SIFT	Deleterious	Deleterious	Deleterious	/	Tolerated	Deleterious	Deleterious
	Polyphen-2	Probably damaging	Probably damaging	Probably damaging	/	Probably damaging	Probably damaging	Probably damaging
	Mutation Taster	Disease causing	Disease causing	Disease causing	/	Disease causing	Disease causing	Disease causing
	CADD [2]	33	29.5	27.5	/	25.4	27.9	34

[1], NM_017539.2; [2], score > 4.0 is predicted to be damaging. /, not applicable.

Asthenoteratozoospermia phenotype is observed in patients with *DNAH3* variants

We next investigated the aberrant sperm morphology of the patients using Papanicolaou staining and SEM analysis. Notably, the tails of sperm from the patients exhibited a typical phenotype associated with MMAF, including coiled, short, bent, irregular, and/or absent flagella (**Figure 2A and B**, **Figure S1A**). In addition, a fraction of defects in the sperm head were also present in the patients' sperm (**Figure 2B**).

TEM was employed to determine the ultrastructure of the sperm from the patients. Compared to the integrated and well-organized "9L+L2" axonemal arrangement of the sperm flagella from the normal control, spermatozoa from the patients showed absent or disordered CPs, MTDs, and outer dense fibers (ODFs) in different regions of the flagella (**Figure 3A**, **Figure S1B**). Interestingly, the IDAs of sperm flagella of the patients were hardly captured compared to the control (**Figure 3A**). Additionally, in the midpiece of sperm flagella of the patients, dissolved mitochondrial material was also observed evidently under TEM (**Figure 3A**). We next conducted immunofluorescence staining to label the mitochondria of patients' sperm with TOM20, a subunit of the mitochondrial import receptor. Remarkably, in contrast to the robust TOM20 signals observed in the normal control, the TOM20 signals in the sperm from the patients were considerably diminished, indicating a disrupted mitochondrial function (**Figure 3B**). Together, these data suggested that *DNAH3* may function in sperm flagellar development, and loss-of-function variants were associated with MMAF in humans.

DNAH3 is exclusively expressed in the sperm flagella of humans and mice

To further understand the function of DNAH3 in male reproduction, we explored the expression pattern of DNAH3 in humans and mice. qPCR results revealed that *Dnah3* was predominantly expressed in the mouse testis (**Figure S2A**). Moreover, when observing the expression of *Dnah3* in testes from mice at different postnatal days, we found that *Dnah3* expression was significantly elevated beginning on postnatal Day 22, peaked at postnatal Day 30, and maintained a stable expression level thereafter (**Figure S2B**). In addition, germ cells at different stages were isolated from the testes of humans and mice and were stained with anti-DNAH3 antibody. The results showed that DNAH3 was expressed in the cytoplasm of spermatocytes and spermatogonia and then obviously in the flagellum of early and late spermatids (**Figure S3A and B**). These expression data suggest that DNAH3 may play an important role in sperm flagellar development during spermatogenesis in humans and mice.

Deletion of *Dnah3* causes male infertility in mice

Considering the absent expression of DNAH3 in the patient sperm, we generated *Dnah3* KO mice using CRISPRLCas9 technology to further confirm the essential role of DNAH3 in spermatogenesis (**Figure S4A**). PCR, qPCR, and immunofluorescence staining were used to confirm that *Dnah3* was null in KO mice (**Figure S4B-E**). The *Dnah3* KO mice survived without any evident abnormalities in development and behavior. H&E staining further revealed that there were no histological differences in the lung, brain, eye, or oviduct between wild-type (WT) and *Dnah3* KO mice (**Figure S5A**). In addition, no obvious abnormalities in ciliary development were observed in these organs in KO mice compared to WT mice (**Figure S5B**). The *Dnah3* KO female mice were fertile with normal oocyte development (**Figure S6A**). However, the *Dnah3* KO male mice were completely infertile (**Figure 4A**). We next examined the testis and epididymis of *Dnah3* KO male mice to elucidate the etiology of infertility. There was no detectable difference in the testis/body weight ratio of *Dnah3* KO mice when compared to WT mice (**Figure S6B**).

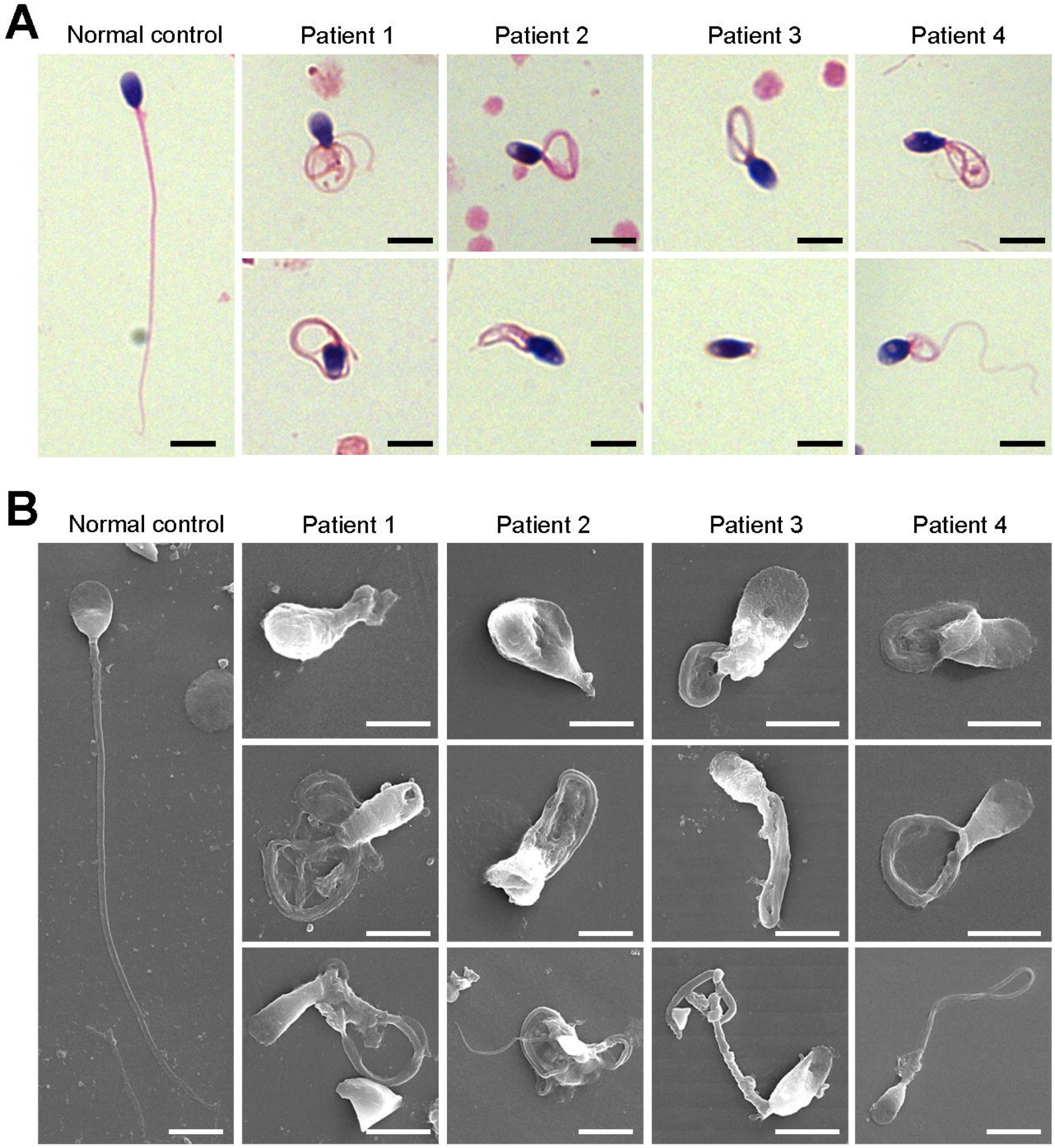


Figure 2.

Defects in sperm morphology of the patients harboring *DNAH3* variants.

(**A**, **B**) Abnormal sperm morphology was observed through Papanicolaou staining (A), and SEM analysis (B) compared to normal control. Scale bars, 5 μm.

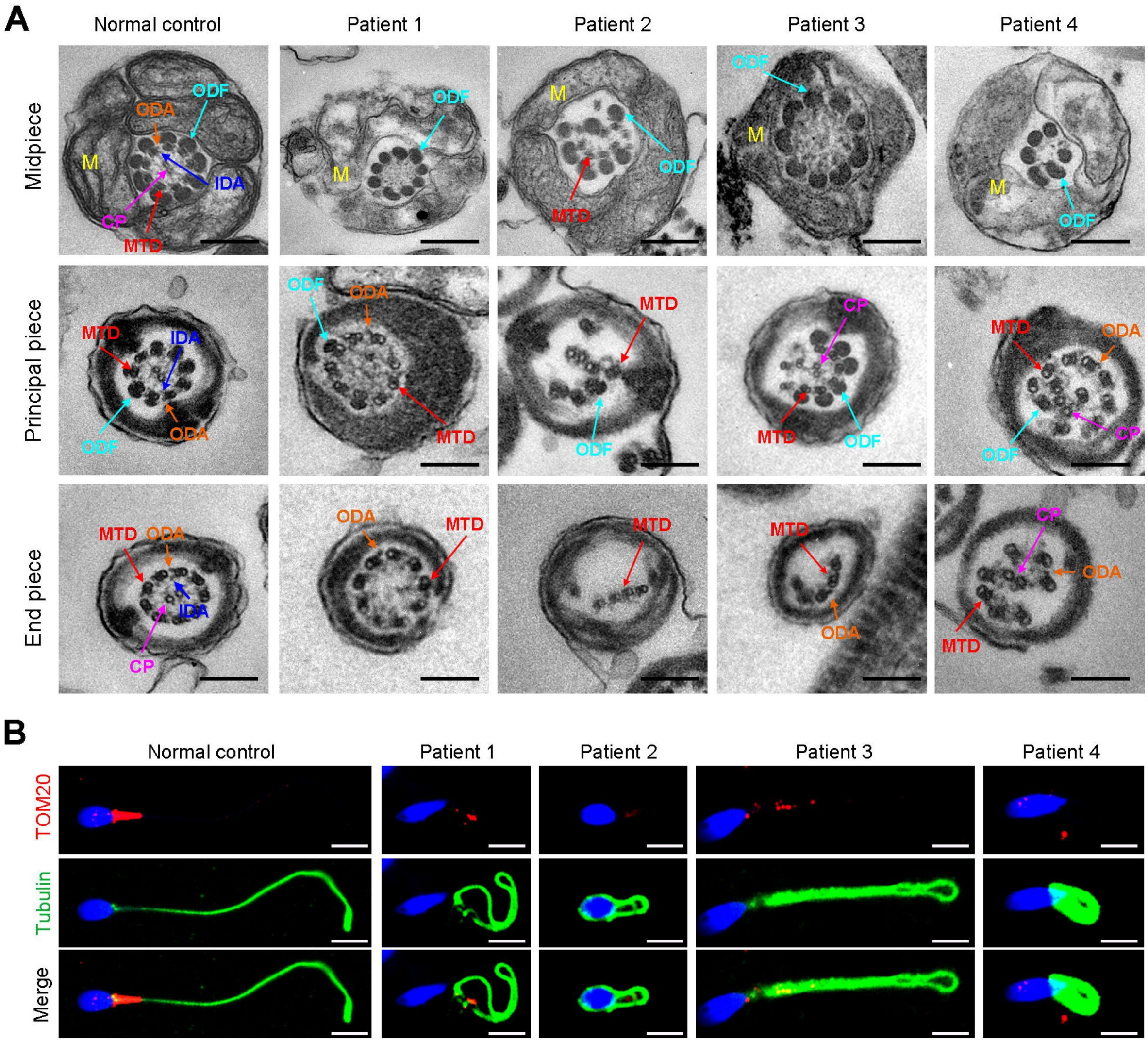


Figure 3.

Ultrastructural and mitochondrial defects in sperm from infertile men with *DNAH3* variants.

(**A**) TEM analysis of sperm obtained from a normal control and patients harboring *DNAH3* variants. Cross-sections of the midpiece, principal piece and endpiece of sperm from normal control showed the typical ''9 + 2'' microtubule structure, and an IDA and an ODA were displayed on the A-tube of each microtubule doublet. Cross-sections of the midpiece, principal piece and endpiece of sperm from the patients displayed absent or disordered CPs, MTDs and ODFs, as well as an evident missing of the IDAs in different pieces of the flagella. M, mitochondria sheath; ODF, outer dense fiber; MTD, microtubule doublets; CP, central pair; IDA, inner dynein arms; ODA, outer dynein arms. Scale bars, 200 nm. (**B**) Immunofluorescence staining of TOM20 in sperm from the patients and normal control. Red, TOM20; green, α-Tubulin; blue, DAPI; scale bars, 5 μm.

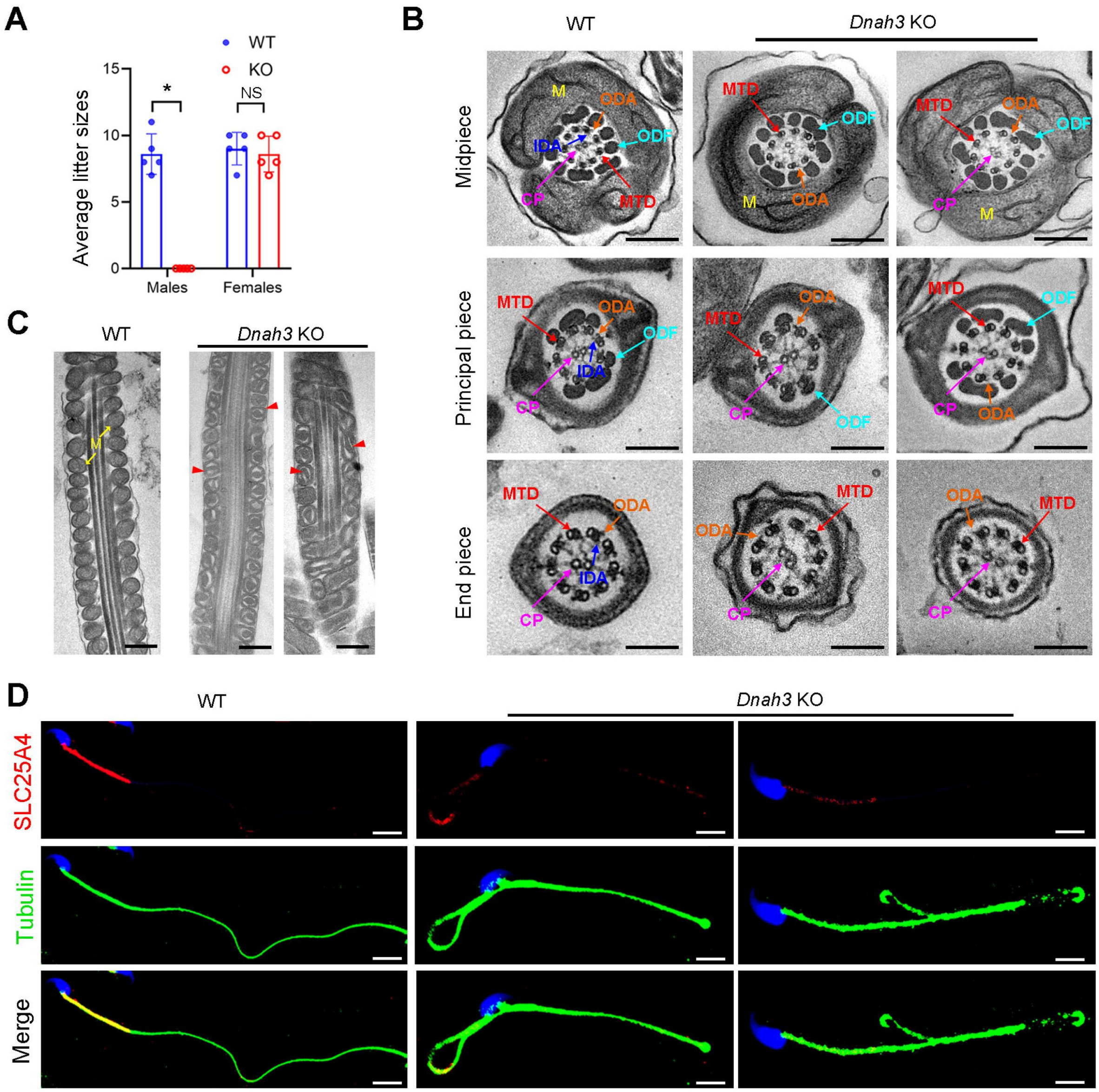


Figure 4.

Dnah3 KO male mice are infertile.

(**A**) Fertility of *Dnah3* KO mice. The KO male mice were infertile (n = five biologically independent WT mice or KO mice; Student's *t* test; *, P < 0.05; NS, no significance; error bars, s.e.m.). (**B**) TEM analysis of the cross-sections of spermatozoa from *Dnah3* KO mice revealed an obvious absence of IDAs in different pieces of the flagella compared to WT mice. M, mitochondrion sheath; ODF, outer dense fiber; MTD, microtubule doublet; CP, central pair; IDA, inner dynein arm; ODA, outer dynein arm. Scale bars, 200 nm. (**C**) Disrupted mitochondria were observed in spermatozoa tail from *Dnah3* KO mice by TEM analysis. The yellow arrows indicate the normal mitochondria. The red arrowheads indicate the dilated intermembrane spaces and dissolved mitochondrial material. M, mitochondrion sheath. Scale bars, 200 nm. (**D**) Immunofluorescence staining of SLC25A4 indicated impaired mitochondrial formation in *Dnah3* KO mice compared to WT mice. Red, SLC25A4; green, α-Tubulin; blue, DAPI; scale bars, 5 μm.

Moreover, subsequent computer-assisted sperm analysis (CASA) also showed that sperm isolated from the cauda epididymis were slightly decreased, and nearly all sperm were completely immobile (**Table 3**, Movie S1 and Movie S2). Papanicolaou staining and SEM analysis revealed morphological defects in partial spermatozoa from *Dnah3* KO mice, including coiled, bent, and irregular flagella, as well as aberrant heads and acephalic spermatozoa (**Figure S7A and B**).

TEM was further utilized to evaluate the sperm flagellar ultrastructure of *Dnah3* KO mice. There were no obvious abnormalities of "9 + 2" microtube arrangement in most sperm from the *Dnah3* KO mice when compared to WT mice (**Figure 4B**, **Figure S7C**). However, in contrast to the clear display of an IDA and an ODA on the A-tube of each microtubule doublet in the sperm flagella of WT mice, the sperm flagella of *Dnah3* KO mice exhibited an absence of almost all the IDAs (**Figure 4B**, **Figure S7D**). In addition, the disrupted mitochondria of spermatozoa from *Dnah3* KO mice were also observed under TEM, as manifested by the dilated intermembrane spaces and dissolved mitochondrial material (**Figure 4B and C**, **Figure S7E**). We next performed immunofluorescence staining to label SLC25A4, which is responsible for the exchange of ATP and ADP across the mitochondrial inner membrane. Strikingly, compared to the bright fluorescence signals in the midpiece of WT sperm, the signals *Dnah3* KO were significantly diminished (**Figure 4D**), indicating impaired mitochondrial function. Collectively, DNAH3 is essential for spermatogenesis, and its deficiency seriously damages the sperm motility and IDAs in both humans and mice.

DNAH3 deficiency impairs IDAs related to the reduction of IDA-associated proteins

Considering the disrupted IDAs revealed by TEM analysis in both our patients and *Dnah3* KO mice, we speculated whether the defective IDAs were attributed to the decreased expression of the key IDA-associated proteins. The immunofluorescence data showed that DNAH1/DNAH6 and DNALI1, corresponding to the heavy and light intermediate chains of the IDAs (16), respectively, were almost invisible along the sperm flagella of the patients when compared to control (**Figure 5A-C**). Consistent results were obtained in our subsequent western blotting analysis of sperm lysates from the patients (**Figure 5D-F**), indicating that DNAH3 may manipulate the assembly of IDA through regulating the expression of IDA-associated proteins. In contrast, DNAH8/DNAH17 and DNAI1, corresponding to the heavy and intermediate chains of ODAs (25), were readily detectable in the patients' sperm flagella and were comparable to the control (**Figure S8A-C**), suggesting that DNAH3 may not regulate the expression of ODA-associated proteins. We also performed immunofluorescence staining and western blotting analysis of DNAH1, DNAH6, DNALI1, DNAH8, DNAH17 and DNAI1 on sperm from *Dnah3* KO mice, and the results observed were consistent with those of the patients (**Figure 6A-F**, **Figure S9A-C**). These findings suggested that other IDA-associated proteins might be downstream effectors of DNAH3, which needs more future research.

ICSI treatment of humans with *DNAH3* variants and *Dnah3* KO mice

ICSI treatment has been reported to be effective in asthenoteratozoospermia-associated infertility (34, 35). ICSI cycles were attempted for the patients after written informed consent was obtained. The female partners all had normal basal hormone levels and underwent a long gonadotrophin-releasing hormone agonist protocol (**Table 4**). The wife of Patient 1 underwent one ICSI attempt. A total of 21 metaphase II (MII) oocytes were retrieved and microinjected, of which 17 oocytes were successfully fertilized (17/21, 80.95%) and cleaved (17/17, 100%). Thirteen Day 3 (D3) embryos were formed, six of which developed into blastocysts (8/13, 61.54%) after standard embryo culture. Two blastocysts were transferred, one of which was implanted. She eventually achieved clinical pregnancy, and the pregnancy is ongoing (**Table 4**). The partner of Patient 2 underwent two ICSI attempts. In her first ICSI attempt, six MII oocytes were retrieved, of which three were fertilized (3/6, 50%) and cleaved (3/3, 100%). After standard embryo culture, two

	WT	KO	P^b value
Semen parameters			
Sperm concentration (10^6 /ml) [a]	112.32 ± 18.26	105.17 ± 11.15	0.059
Motility (%) [b]	71.56 ± 3.97	4.37 ± 1.15	<0.01
Progressive motility (%) [b]	60.36 ± 4.32	4.37 ± 1.15	<0.01
Sperm locomotion parameters			
Curvilinear velocity (VCL) (μm/s) [b]	67.54 ± 6.79	9.07 ± 1.22	<0.01
Straight-line velocity (VSL) (μm/s) [b]	28.91 ± 4.86	2.68 ± 0.52	<0.01
Average path velocity (VAP) (μm/s) [b]	39.02 ± 5.31	3.85 ± 0.82	<0.01
Amplitude of lateral head displacement (ALH) (μm) [b]	0.71 ± 0.03	0.13 ± 0.04	<0.01
Linearity (LIN) [b]	0.43 ± 0.07	0.30 ± 0.02	0.037
Wobble (WOB, = VAP/VCL) [b]	0.58 ± 0.06	0.42 ± 0.05	0.024
Straightness (STR, = VSL/VAP)	0.74 ± 0.22	0.70 ± 0.13	0.80
Beat-cross frequency (BCF) (Hz) [b]	4.86 ± 0.12	0.73 ± 0.08	<0.01

[a] Epididymides and vas deferens.

[b] A significant difference, two-sided student's t-test. n = 3 biologically independent WT mice or KO mice.

Table 3.

Semen analysis using CASA in the *Dnah3* KO mice.

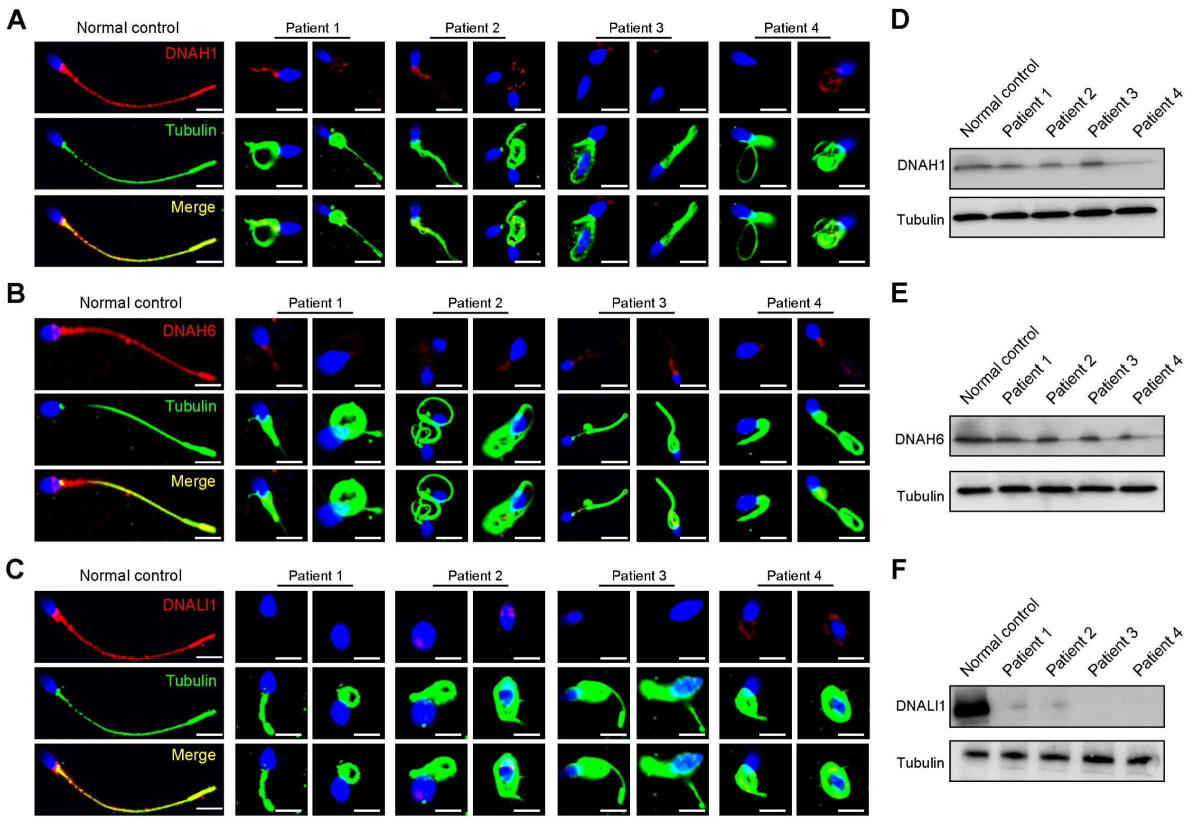


Figure 5.

Immunofluorescence staining and western blotting analysis of IDA-associated proteins in spermatozoa obtained from normal control and patients with *DNAH3* variants.

(**A** – **C**) Immunofluorescence staining of DNAH1 (A), DNAH6 (B) and DNALI1 (C) in spermatozoa from patients and normal controls. Red, DNAH1 in (A), DNAH6 in (B), DNALI1 in (C); green, α-Tubulin; blue, DAPI; scale bars, 5 μm. (**D** – **F**) Western blotting analysis of DNAH1(D), DNAH6 (E), DNALI1 (F) in sperm lysates from the patients and normal control.

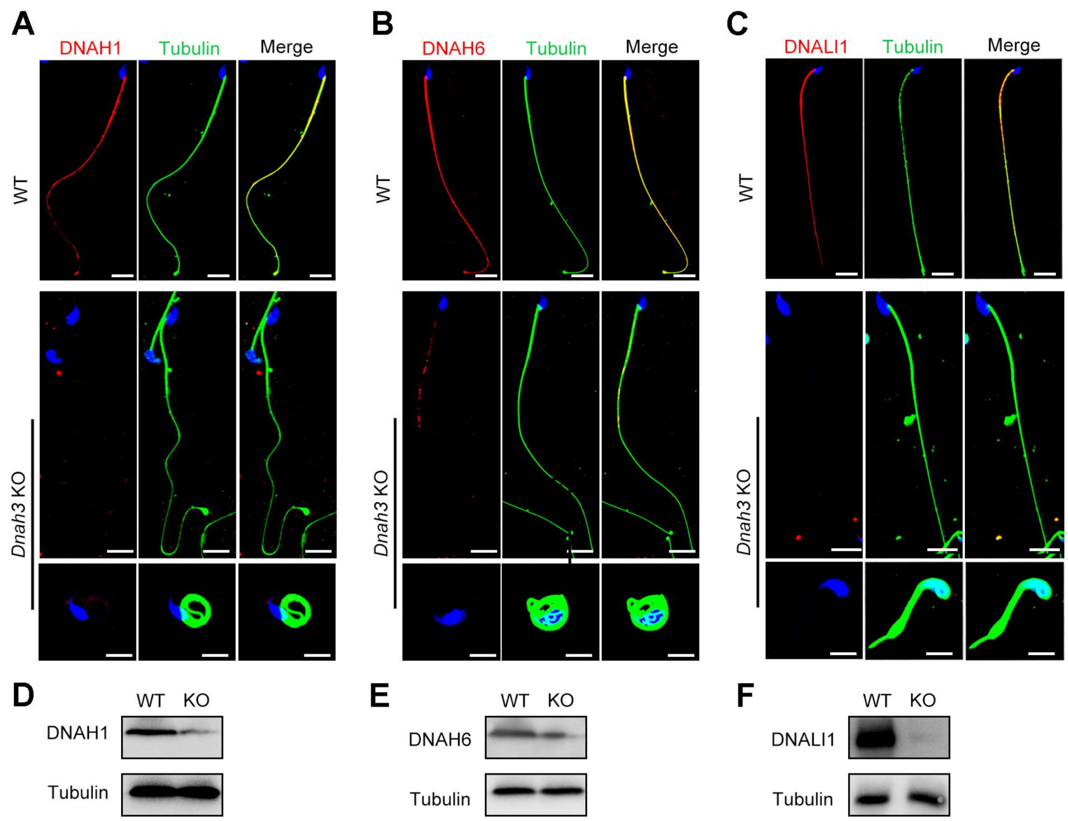


Figure 6.

Immunofluorescence staining and western blotting analysis of IDA-associated proteins in spermatozoa from WT and *Dnah3* KO mice.

(**A** – **C**) Immunofluorescence staining of DNAH1 (A), DNAH6 (B) and DNALI1 (C) in spermatozoa from *Dnah3* KO and WT mice. Red, DNAH1 in (A), DNAH6 in (B), DNALI1 in (C); green, α-Tubulin; blue, DAPI; scale bars, 5 μm. (**D** – **F**) Western blotting analysis of DNAH1(D), DNAH6 (E) and DNALI1 (F) in spermatozoa lysates from *Dnah3* KO and WT mice.

D3 embryos were formed and transferred. However, this ICSI failed because no embryos were implanted. In her second ICSI attempt, all five MII oocytes were fertilized and cleaved (5/5, 100%). Five D3 embryos were obtained, of which two were transferred, but no embryos were implanted. The remaining three D3 embryos were cultured continuously, and two available blastocysts were formed and kept to be transferred in the future (**Table 4**). The partner of Patient 3 underwent one ICSI attempt. Of the 20 MII oocytes retrieved, 19 oocytes were fertilized (19/20, 95.0%) and cleaved (19/19, 100%). Fifteen D3 embryos were obtained, and 10 developed into available blastocysts (10/15, 66.7%). One blastocyst was transferred and implanted. She achieved clinical pregnancy, and the pregnancy is ongoing (**Table 4**). The wife of Patient 4 underwent four failed ICSI attempts. In her first two ICSI attempts, 13 and 12 MII oocytes were retrieved, of which five (5/12, 41.67%) and six (6/13, 46.15%), respectively, were fertilized and cleaved. Two available D3 embryos were obtained and transferred in both ICSI attempts, but no embryos were implanted. In her third ICSI attempt, of the eight MII oocytes retrieved, four (4/8, 50%) were fertilized and cleaved (4/4, 100%). However, no available D3 embryos were acquired. In her last ICSI attempt, seven MII oocytes were retrieved, of which three were fertilized (3/7, 42.68%) and two were cleaved (2/3, 66.7%), but no available D3 embryos were formed (**Table 4**). The vivid embryonic development of the partner of Patient 1 and Patient 3 after ICSI treatment was shown in **Figure 7A**.

We also carried out ICSI treatment on *Dnah3* KO male mice. Strikingly, favorable outcomes of ICSI were obtained in *Dnah3* KO male mice. After injection of spermatozoa from *Dnah3* KO male mice, pronuclei were observed in most embryos in both the KO and WT groups, indicating a normal fertilization rate (**Figure 7B**). There was no difference in the percentage of 2-cell and blastocyst-stage embryos between the KO and WT groups (**Figure 7B**). Collectively, we observed successful ICSI outcomes in two out of four DNAH3-deficient patients and *Dnah3* KO male mice and therefore suggested ICSI as an optional treatment for infertile men harboring biallelic pathogenic variants in *DNAH3*, and the additional female risk factors for infertility should not be excluded in the failed patients.

Discussion

In the present study, we identified pathogenic variants in *DNAH3* in unrelated infertile men with asthenoteratozoospermia. These variations resulted in the almost absence of DNAH3 and sharply decreased the expression of other IDA-associated proteins, including DNAH1, DNAH6 and DNALI1. Combined with similar findings in *Dnah3* KO mice, we demonstrated that DNAH3 is fundamental for male fertility. Moreover, we suggest that ICSI might be a favorable treatment for male infertility caused by DNAH3 deficiency. Our findings identify a function for DNAH3 in male reproduction in humans and mice and may provide a new view on the clinical practice of male infertility.

Recently, Meng et al. identified bi-allelic variants in *DNAH3* from patients diagnosed with asthenoteratozoospermia, revealing multiple morphological defects and a disrupted “9+2” arrangement in the patients’ sperm (36). Furthermore, they generated *Dnah3* KO mice, which were infertile, and exhibited moderate morphological abnormalities with a normally structured “9 + 2” microtubule arrangement (36). In our study, we also observed similar phenotypic differences between the phenotypes of *DNAH3*-deficient patients and *Dnah3* KO mice. These findings indicate that DNAH3 may play crucial yet distinct roles in human and mouse male reproduction. Additionally, our TEM analysis demonstrated a notable absence of IDAs in sperm from both *DNAH3*-deficent patients and *Dnah3* KO mice, resembling the findings of Meng et al. To further investigate, we conducted immunofluorescent staining and western blotting to assess the levels of IDA-associated proteins (DNAH1, DNAH6 and DNALI1) and ODA-associated proteins (DNAH8, DNAH17 and DNAI1) in sperm samples from both our *DNAH3*-deficient patients and *Dnah3* KO mice. Our data revealed a reduction in IDA-associated protein levels and comparable

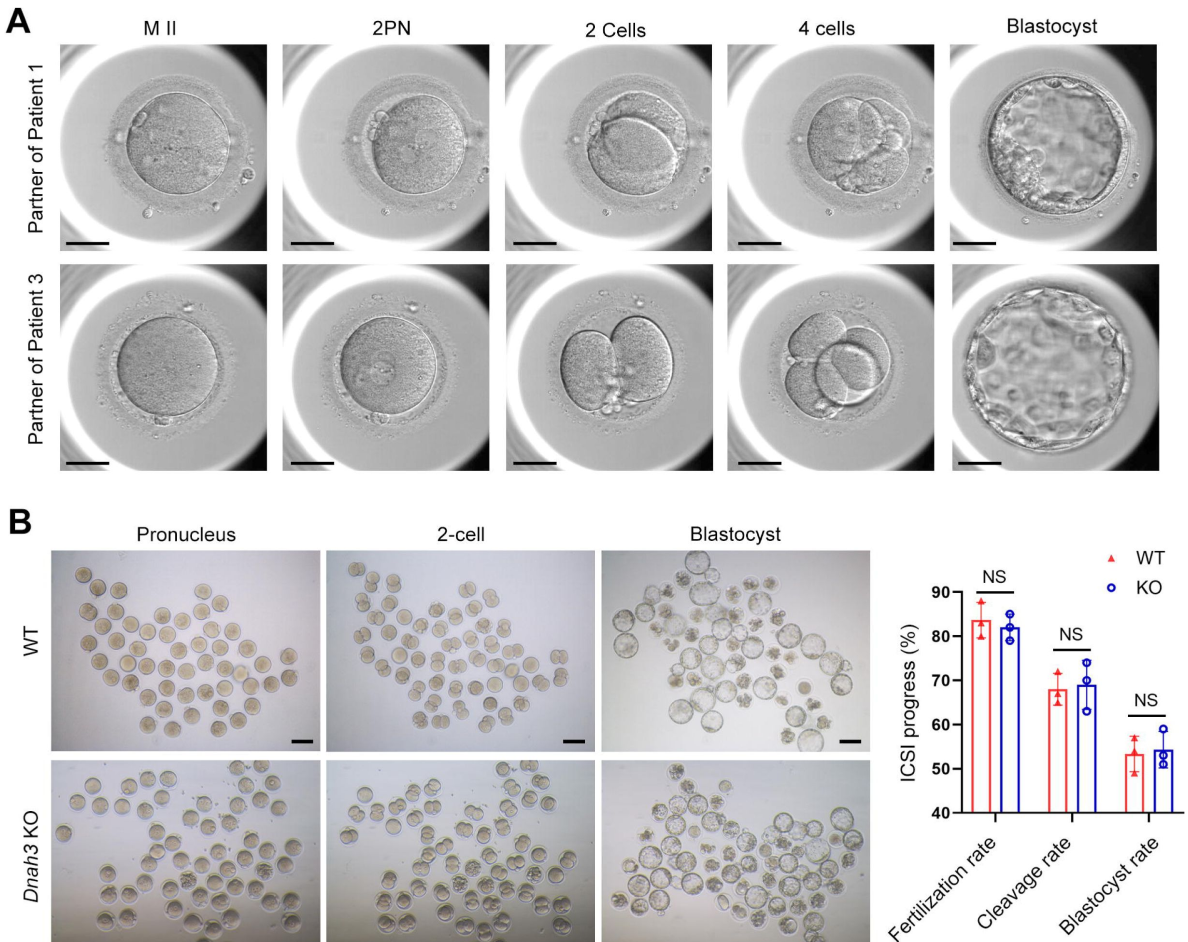


Figure 7.

ICSI outcomes of *DNAH3*-deficient patients and *Dnah3* KO mice.

(**A**) The embryonic development of Patient 1 and Patient 3 after ICSI treatment. MII, metaphase II; PN, pronucleus; scale bars, 40 μm. (**B**) There was no difference in the fertilization rate or 2-cell and blastocyst embryo formation rates between the *Dnah3* KO and WT groups (n = three biologically independent WT mice or KO mice; Student's *t* test; NS, no significance; error bars, s.e.m.).

	Patient	Patient 1	Patient 2		Patient 3	Patient 4			
Subjects	Female age (y)	24	30		30	36			
	Length of primary infertility history (y)	6	1		3	8			
Basal hormones	FSH (IU/L)	7.1	8.5		3.49	4.3			
	LH (IU/L)	4.44	2.5		4.2	2.6			
	E2 (pg/mL)	83	50		43.49	68			
	Prog (ng/mL)	0.2	0.6		0.3	0.3			
ICSI Cycles		Cycle 1	Cycle 1	Cycle 2	Cycle 1	Cycle 1	Cycle 2	Cycle 3	Cycle 4
	E2 level on the trigger day (pg/ml)	3366	1519.6	1582.4	>5000	4440	5000	3152	2206
	No. of follicles ≥ 14 mm on the trigger day	15	6	5	20	13	12	14	11
	No. of follicles ≥ 18 mm on the trigger day	10	3	4	8	11	9	7	5
	No. of oocytes retrieved	24	6	5	25	16	20	21	14
ICSI progress	No. of MII oocytes	21	6	5	20	12	13	8	7
	Fertilization rate (%)	80.95 17/21)	50 (3/6)	100 (5/5)	95 (19/20)	41.67 (5/12)	46.15 (6/13)	50 (4/8)	42.86 (3/7)
	Cleavage Rate (%)	100 (17/17)	100 (2/2)	100 (5/5)	100 (19/19)	100 (5/5)	100 (6/6)	100 (4/4)	66.7 (2/3)
	Available D3 embryos	13	2	5	15	2	2	0	0
	Blastocyst formation rate (%)	61.5 (8/13)	0	66.7 (2/3)	66.7 (10/15)	-	-	-	-
Clinical outcomes	No. of embryos transferred	2 blastocysts	0	2 D3	1 blastocyst	2 D3	2 D3	-	-
	Implantation rate (%)	50 (1/2)	0	0	100 (1/1)	0	0	-	-
	Clinical pregnancy	Yes	No	No	Yes	-	-	-	-
	No. of live birth	ongoing	-	-	ongoing	-	-	-	-

-, not applicable.

Table 4.

Outcomes of ICSI treatment in the patients with *DNAH3* mutations.

ODA-associated protein levels in comparison to normal controls and WT mice, respectively, thus corroborating the TEM observations. These results suggest that DNAH3 is involved in sperm flagellar development in human and mice, specifically through its role in the assembly of IDAs.

Primary ciliary dyskinesia (PCD, MIM: 244400) is a genetic disorder affecting at least one in 7554 individuals (37). The most common symptoms of PCD are recurrent infections in airways due to malfunction of the motile cilia that are responsible for mucus clearance (38). It has been suggested that male infertility associated with sperm defects is highly prevalent (up to 75%) among individuals with PCD (39). Axonemal defects caused by variants within *DNAH* family members, including *DNAH5*, *DNAH6*, *DNAH7*, *DNAH9* and *DNAH11*, are causative factors for PCD (40–42). Moreover, deficiency in these PCD-causing *DNAH*s has also been associated with male infertility (9, 14, 20, 21, 43–45). Additionally, other *DNAH*s, such as *DNAH1*, *DNAH2*, *DNAH8*, *DNAH10*, *DNAH12* and *DNAH17*, are suggested to be pathogenic genes of isolated male infertility (18, 19, 22, 24, 25). These phenotype–genotype correlations may be attributed to the fact that ciliary and flagellar axonemes have cell type-specific or cell type-enriched DNAHs (46). DNAH3 resides in the IDA and is expressed in testis and ciliary tissues, including the lung, brain, eye, and oviduct. However, despite its presence in these tissues, the relationship between deficient DNAH3 and disease is unclear to date. Intriguingly, in our study, none of the patients with *DNAH3* deficiency reported experiencing any of the principal symptoms associated with PCD. Additionally, our *Dnah3* KO mice exhibited normal ciliary development in the lung, brain, eye, and oviduct. Similarly, Meng et al. did not mention any PCD symptoms in their *DNAH3*-deficient patients, and their *Dnah3* KO mice also demonstrated normal ciliary morphology in the trachea and brain (36). These combined observations suggest that DNAH3 may play a more important role in sperm flagellar development than in other motile cilia functions. Given that DNAH3 is expressed in ciliary tissues, its role in these tissues remains intriguing and could be elucidated through sequencing of larger cohorts of individuals with PCD.

ICSI has been an efficient treatment for male infertility (47, 48). However, the outcomes of ICSI for male infertility caused by variants in different *DNAH* genes are variable. It has been demonstrated that infertile males with variants in *DNAH1*, *DNAH2*, *DNAH7*, and *DNAH8* have a favorable prognosis (22, 49–53), while patients with variants in *DNAH17* have poor outcomes after ICSI treatment (25, 54). Meanwhile, the ICSI outcomes in male infertility caused by *DNAH6* variants may depend on the specific mutation or be controversial (20, 55, 56). The patients with *DNAH3* mutations in our study experienced different clinical outcomes of ICSI treatment. The partners of Patient 1 and Patient 3 achieved clinical pregnancy. The wives of Patient 2 and Patient 4 obtained favorable fertilization and cleavage rates but experienced no clinical pregnancy due to the nonimplantation of the transferred embryos. Remarkably, despite the diverse variants within *DNAH3* observed in the four patients, all variants led to a complete absence of DNAH3 expression. Additionally, we did not identify any pathogenic variants that associated with fertilization failure and early embryonic development in the two patients with failed ICSI outcomes. Therefore, these different ICSI outcomes might be attributed to additional unexplained factors from the female partners. Importantly, in the study from Meng et al., one patient carrying *DNAH3* variants received ICSI treatment, and the partner obtained clinical pregnancy (36). Combined with the successful ICSI outcomes observed in *Dnah3* KO mice, we suggest ICSI as an optimized treatment for infertile men carrying variants in *DNAH3*. More cases are needed to precisely estimate the prevalence of *DNAH3* mutations and determine a prognosis for ICSI treatments.

In conclusion, our study revealed an unexplored role of DNAH3 in male reproduction in humans and mice, suggesting *DNAH3* as a novel causative gene for human asthenoteratozoospermia. Moreover, ICSI is as an optimized treatment for infertile men with *DNAH3* variants. This study expands our knowledge of the relationship between DNAH proteins and disease, facilitating genetic counseling and clinical treatment of male infertility in the future.

Methods

Human subjects

Four unrelated Han Chinese infertile men and their family members were recruited from West China Second University Hospital of Sichuan University and Women and Children's Hospital of Chongqing Medical University. All patients exhibited a normal karyotype (46 XY) without deletion of the azoospermia factor (AZF) region in the Y-chromosome. All of the participants were provided informed consent, and the study was approved by the ethics committee of West China Second University Hospital and The First Affiliated Hospital of Chongqing Medical University.

Genetic analysis

Peripheral blood samples were obtained from the subjects to extract genomic DNA using a DNA purification kit (TIANGEN, DP304). For WES, 1 µg of genomic DNA was utilized for exon capture using the Agilent SureSelect Human All Exon V6 Kit and sequenced on the Illumina HiSeq X system (150-bp read length). The quality of WES, including clean reads, sequencing depth, sequencing coverage, and mapping quality are listed in Table S1. The variants identified through WES were annotated and filtered using Exomiser. Next, the variants were screened to obtain candidate variants based on the following criteria: (1) the allele frequency in the East Asian population was less than 1% in any database, including the ExAC Browser, gnomAD, and the 1000 Genomes Project; (2) the variants affected coding exons or canonical splice sites; (3) the variants were predicted to be possibly pathogenic or damaging. The remain genes were then analyzed using the Human Protein Atlas (HPA) database (*https://www.proteinatlas.org/*) and Mouse Genome Informatics (MGI) database (*https://informatics.jax.org/*) to access their expression in human and mouse testis. Additionally, OMIM database (*https://www.omim.org/*) and relevant literature were used to understand their relationship with human infertility. Given the assumption of a recessive inheritance pattern, monoallelic variants were excluded from consideration. The remained candidate pathogenic variants were verified by Sanger sequencing on DNA from the patients' families. The primer pairs used for PCR amplification are listed in Table S2.

Electron microscopy

For scanning electron microscopy (SEM), sperm samples were fixed in glutaraldehyde (2.5%, w/v) and dehydrated using an ethanol gradient (30, 50, 75, 85, 95, and 100% ethanol). The samples were dried using a CO_2 critical-point dryer (Eiko HCP-2, Hitachi) and observed under SEM (S-3400, Hitachi). For transmission electron microscopy (TEM), sperm samples were fixed in glutaraldehyde (3%, w/v) and osmium tetroxide (1%, w/v) and dehydrated with an ethanol gradient. The samples were embedded in Epon 812. Ultrathin sections were stained with uranyl acetate and lead citrate and analyzed under TEM (Tecnai G2 F20).

STA-PUT velocity sedimentation

Single testicular cells from obstructive azoospermia and 8-week-old C57BL male mice were obtained using the STA-PUT velocity sedimentation method as described previously (57, 58). In brief, total spermatogenic cells were harvested by digesting seminiferous tubules with collagenase (Invitrogen, 17100017), trypsin (Sigma, T4799) and DNase (Promega, M6101) for 15 min each at 37 °C. Cells were diluted in bovine serum albumin (BSA, 3%, w/v) and filtered through an 80 mm mesh to remove fragments. Then, the cells were resuspended in BSA (3%, w/v) and loaded into an STA-PUT velocity sedimentation cell separator (ProScience) to obtain germ cells at different stages.

RNA isolation and quantitative PCR (qPCR)

Total RNA of mouse tissues was extracted using TRIzol reagent (Invitrogen,15596026,) and reverse-transcribed using the 1st Strand cDNA Synthesis Kit (Yeasen, HB210629) according to the manufacturer's instructions. qPCR was carried out on an iCycler RTLPCR Detection System (Bio-Rad Laboratories) using SYBR Green qPCR Master Mix (Bimake, B21202). Primer sequences are listed in Table S2.

Immunofluorescence staining

Sperm samples were fixed in paraformaldehyde (4%, w/v), permeabilized with Triton X-100 (0.3% v/v) and blocked with BSA (3%, w/v) at room temperature. Samples were incubated with primary antibodies, including DNAH1 (Cusabio, CSB-PA878961LA01HU, 1:100), DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:100), DNAH6 (Proteintech, 18080-1-AP, RRID: AB_2878493, 1:50), DNAH8 (Atlas, HPA028447, RRID: AB_10599600, 1:200), DNAH17 (Proteintech, 24488-1-AP, RRID: AB_2879568, 1:50), DNAI1 (Proteintech, 12756-1-AP, RRID: AB_10643244, 1:50), DNALI1 (Proteintech, 17601-1-AP, RRID: AB_2095372, 1:50), TOM20 (Proteintech, 11802-1-AP, RRID: AB_2207530, 1:50), SLC25A4 (Signalway, 32484, RRID: AB_2941094, 1:100), lectin PNA (Invitrogen, L-32460, 1:50) and alpha tubulin (Abcam, ab7291, RRID: AB_2241126, 1:500), overnight at 4 °C. The next day, the samples were washed and incubated with the secondary antibody Alexa Fluor 488 (Invitrogen, A11008, RRID: AB_143165, 1:1000) or Alexa Fluor 594 (Invitrogen, A11005, RRID: AB_141372, 1:1000), and the nuclei were labeled with 4',6-diamidino-2-phenylindole (DAPI, SigmaLAldrich, D9542). Image capture was performed by a laser scanning confocal microscope (Olympus, FV3000).

For staining of mouse tissues, samples were first fixed in paraformaldehyde (4%, w/v) and dehydrated with an ethanol gradient. Then, the samples were embedded in paraffin and sliced into 5-μm sections. After deparaffinization and rehydration, sections were processed with 3% hydrogen peroxide and incubated in sodium citrate for antigen repair. Subsequently, sections were blocked with goat serum and incubated with primary antibodies against DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:100) or ac-Tubulin (Abcam, ab24610, RRID: AB_448182, 1:500) at 4 °C overnight. The next day, the sections were incubated with the secondary antibody Alexa Fluor 488, followed by labeling the nuclei with DAPI. Image capture was performed using a fluorescence microscope (Zeiss, Ax10).

Western Blotting

Sperm samples were lysed in RIPA buffer (Beyotime, P0013B) to extract the total protein. For analysis of DNALI1, the protein samples were mixed with SDS loading buffer (P0015, Beyotime, China), boiled at 95 °C for 5 minutes, and separated by 12.5% SDS-PAGE. For analysis of DNAH1, DNAH3, DNAH6, the protein samples were mixed with NuPAGE™ LDS sample buffer (Invitrogen, NP0007), denatured at 70°C for 10 minutes, and separated by 3–8% NuPAGE™ Tris-Acetate gels (EA0375BOX, Invitrogen). Then the resolved proteins were transferred to 0.45 μm PVDF membranes (Merck Millipore, IPVH00010). The membranes were blocked, incubated with primary antibodies, including DNALI1 (Proteintech, 17601-1-AP, RRID: AB_2095372, 1:150), DNAH3 (Cusabio, CSB-PA823461LA01HU, 1:200), DNAH6 (Proteintech, 18080-1-AP, RRID: AB_2878493, 1:150) and alpha tubulin (Proteintech, 11224-1-AP, RRID: AB_2210206, 1:1000) at 4 L overnight. The following day, membranes were washed and incubated with HRP-conjugated secondary antibody (Proteintech, SA00001-2, RRID: AB_2722564, 1:5000). Protein bands were visualized using enhanced chemiluminescence reagents (Millipore, WBKLS0500).

Histology hematoxylin-eosin (H&E) staining

Tissue samples from mice were fixed with 4% paraformaldehyde (w/v) overnight. Following dehydration by ethanol, the samples were embedded in paraffin and sliced into 5-μm sections. The sections were stained with hematoxylin and eosin and observed under a microscope (Zeiss, Axio Imager 2).

Generation of the *Dnah3* KO mouse model

Animal experiments in this study were approved by the Experimental Animal Management and Ethics Committee of West China Second University Hospital, Sichuan University, and complied with the Animal Care and Use Committee of Sichuan University. A *Dnah3* knockout mouse model was generated by the CRISPRLCas9 system. Briefly, Cas9 and signal-guide RNAs (5'-GTATCAAGTGGATGTAAACC-3') were transcribed using T7 RNA polymerase in vitro and comicroinjected into the cytoplasm of single-cell C57BL/6J mouse embryos to generate frameshift mutations by nonhomologous recombination through introduction of a 1 bp insertion in exon 13. Then, the embryos were cultured and transferred into the oviducts of pseudopregnant female mice at 0.5 days post-coitum. A mutation of *Dnah3* in the founder mouse and their offspring was confirmed using PCR and Sanger sequencing. The primers used for the generation of animal models are listed in Table S2.

Intracytoplasmic sperm injection (ICSI)

ICSI was carried out using standard techniques. In brief, one-month-old female KM mice were injected with 5 IU of equine chorionic gonadotropin (eCG) (ProSpec, HOR-272) to induce superovulation. Metaphase II-arrested (MII) oocytes were acquired through another injection of 5 IU human chorionic gonadotropin after 48 hours. MII oocytes were incubated with Chatot-Ziomek-Bavister medium (Easycheck, M2750) at 37.5 °C and 5% CO_2 until use. Mouse cauda epididymal spermatozoa were incubated in human tubal fluid (HTF) medium (Easycheck, M1150) and then frozen and thawed repeatedly to remove sperm tails. For ICSI, a single sperm head was microinjected into an MII oocyte by using a NIKON inverted microscope and a Piezo (PrimeTech, Osaka, Japan) in Whitten's-HEPES medium containing 0.01% polyvinyl alcohol (Gibco,12360-038) and cytochalasin B (3.5 g/ml; SigmaLAldrich, C-6762). The successfully injected oocytes were transferred into G1-Plus medium (Vitrolife, 10132) and incubated at 37.5 °C and 5% CO_2. The animal experiments were approved by the Experimental Animal Management and Ethics Committee of West China Second University Hospital, Sichuan University.

Statistical analysis

Prism (version 8.4.0, GraphPad, Boston, MA, USA) and SPSS (version 18.0, IBM Corporation, Armonk, NY, USA) were used to perform statistical analyses. All data are presented as the means ± SEMs. Data from two groups were compared using an unpaired, parametric, two-sided Student's t test, and a p value less than 0.05 was considered statistically significant.

Ethics approval

This study was approved by Ethical Review Board of West China Second University Hospital, Sichuan University. Informed consent was obtained from each participate in this study before taking part.

Data availability

The published article includes all datasets generated or analyzed during this study. The whole exome-sequencing data were deposited in the National Genomics Data Center (NGDC) (*https://ngdc.cncb.ac.cn/*, accession number: HRA007467).

Competing interests

The authors declare that they have no conflict of interest.

Author contributions

Xiang Wang: Data curation; formal analysis; investigation; methodology; writing – original draft. **Gan Shen**: Data curation; formal analysis; **Yihong Yang**: Resources; investigation; methodology. **Chuan Jiang**: Data curation; formal analysis. **Tiechao Ruan**: Formal analysis; methodology. **Xue Yang**: Validation; investigation. **Liangchai Zhuo**: Investigation; **Yingteng Zhang**: Investigation, methodology. **Yangdi Ou**: Investigation. **Xinya zhao**: Investigation. **Shunhua Long**: Methodology. **Xiangrong Tang**: Investigation. **Tingting Lin**: Conceptualization; funding acquisition; project administration. **Ying Shen**: Conceptualization; supervision; project administration; writing – review and editing.

Acknowledgements

The authors thank the patient and his family members for their voluntary participation. We are grateful to Guiping Yuan from Analytical and Testing Center of Sichuan University and Yan Liang from Research Core Facility of West China Hospital, Sichuan University for their help with TEM images and preparing histology slides. This work was supported by National Natural Science Foundation of China (82301807).

Supporting information

Figure S1

Statistics analyzing of aberrant sperm morphology and axonemal ultrastructure observed in *DNAH3*-deficient patients.

(**A**) The percentage of abnormal flagellar morphology in the normal control and patients. (**B**) The percentage of aberrant ultrastructure in different cross-sections of sperm from the normal control and patients.

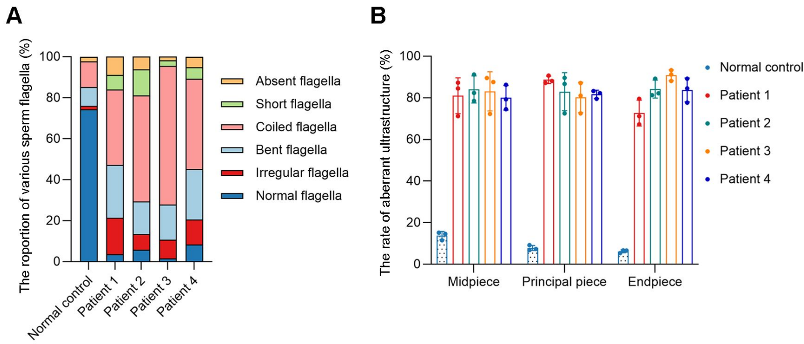
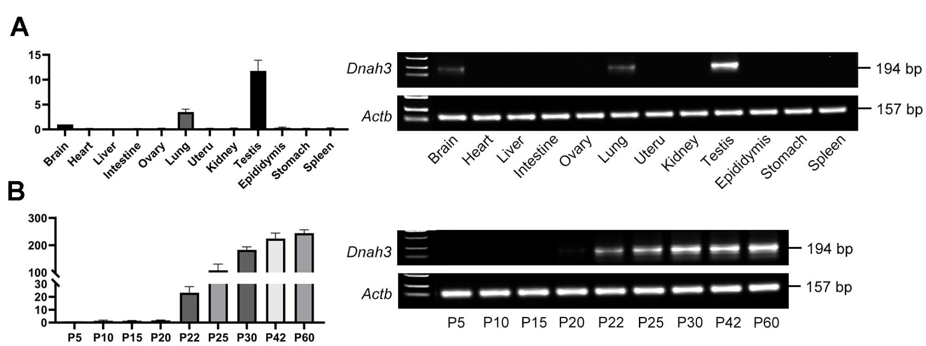


Figure S2

The expression of *DNAH3* in mouse testis.

(**A**) qPCR analysis revealed that *Dnah3* was highly expressed in the mouse testis. (**B**) qPCR analysis showed that *Dnah3* expression was significantly elevated beginning on postnatal Day 12, peaked at postnatal Day 30, and maintained a stable expression level thereafter.



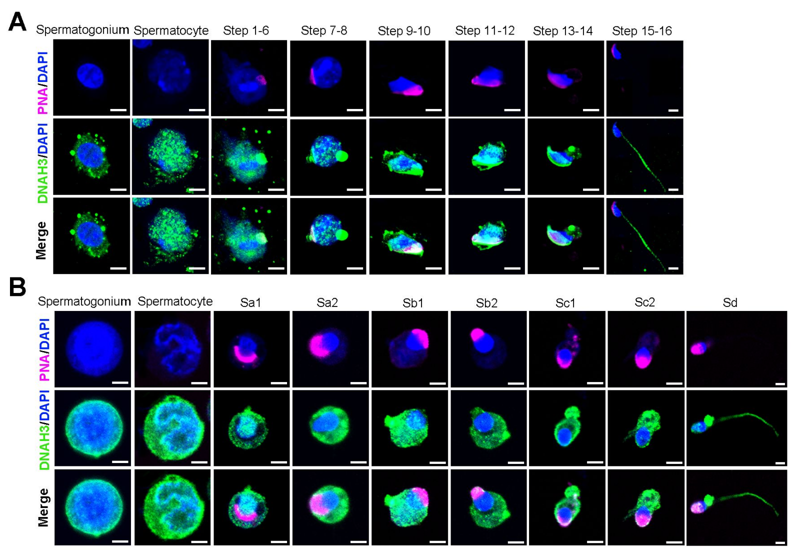


Figure S3

DNAH3 is expressed during spermatogenesis in mice and humans.

(**A**) Immunofluorescence staining of DNAH3 in isolated mouse germ cells. Pink, PNA; green, DNAH3; blue, DAPI; scale bars, 5 μm. (**B**) Immunofluorescence staining of DNAH3 in isolated human germ cells. Pink, PNA; green, DNAH3; blue, DAPI; scale bars, 5 μm.

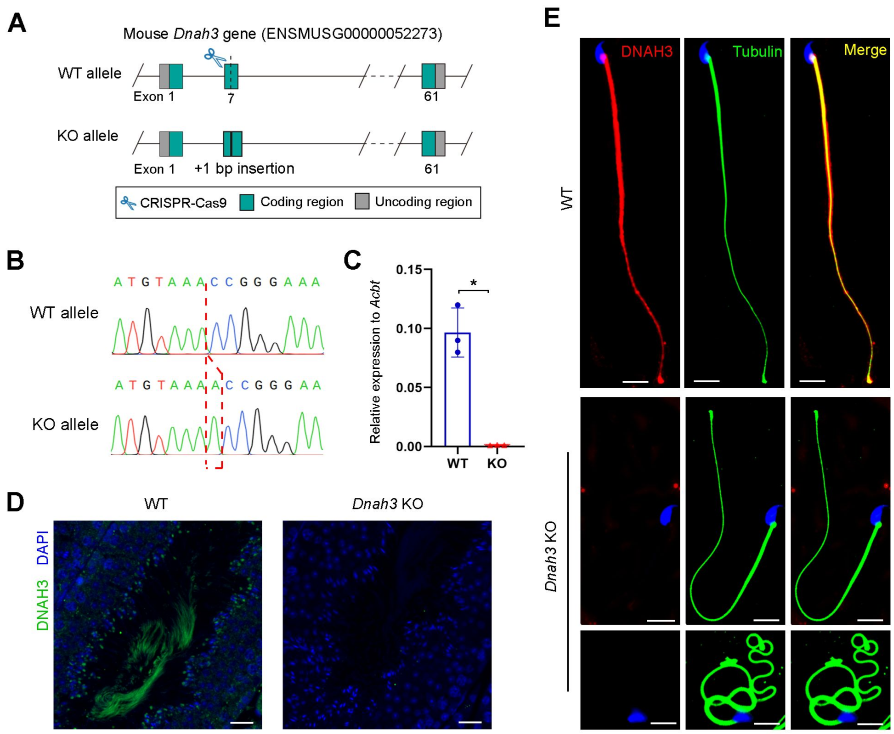


Figure S4

Generation of *Dnah3* KO mice.

(**A**) Schematic illustration of the strategy for the generation of *Dnah3* KO mice. (**B, C**) PCR sequencing (B) and qPCR (C) were used to confirm the genotype and KO efficiency (n = three biologically independent WT mice or KO mice; Student's *t* test; *, P<0.05; error bars, s.e.m.). (**D**) Immunofluorescence staining of DNAH3 in testis of *Dnah3* KO mice and WT mice. Green, DNAH3; blue, DAPI; scale bars, 75 μm. (**E**) Immunofluorescence staining of DNAH3 in spermatozoa isolated from the cauda epididymis of *Dnah3* KO mice and WT mice. Red, DNAH3; green, α-Tubulin; blue, DAPI; scale bars, 5 μm.

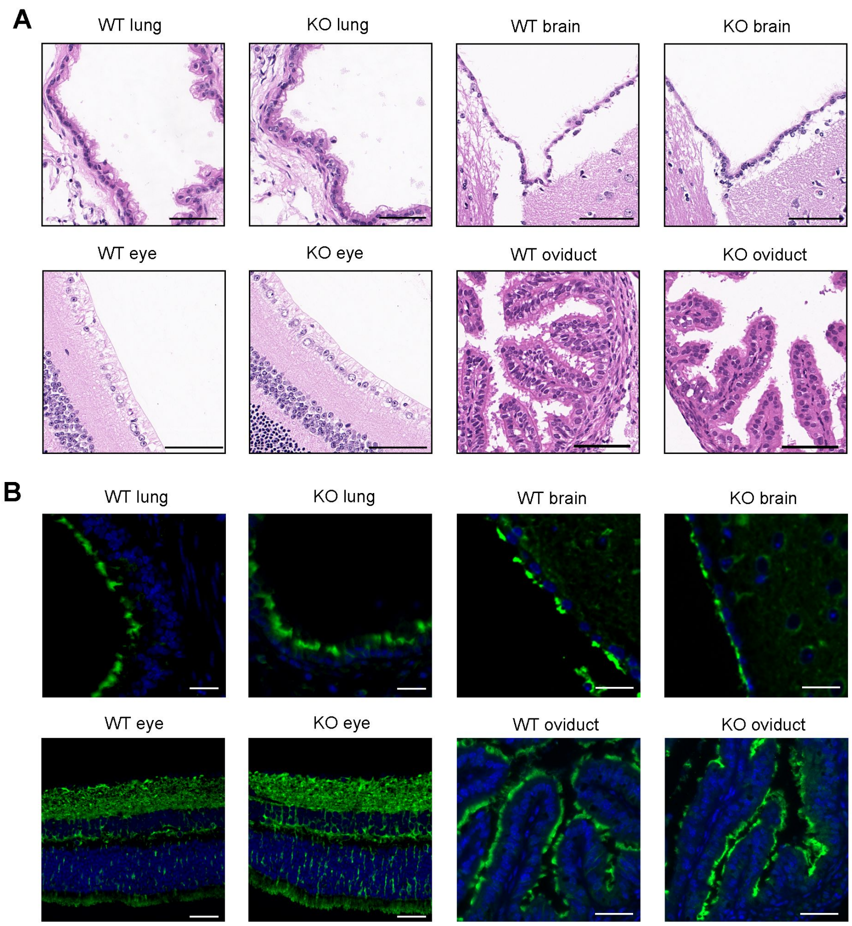


Figure S5

Ciliary development of *Dnah3* KO mice.

(**A**) H&E staining of lung, brain, eye, and oviduct from *Dnah3* KO mice and WT mice. Scale bars, 100 μm. (**B**) Analysis of ciliary development in the lung, brain, eye, and oviduct from *Dnah3* KO mice and WT mice by using immunofluorescence staining. Green, Ac-Tubulin; blue, DAPI; scale bars, 20 μm.

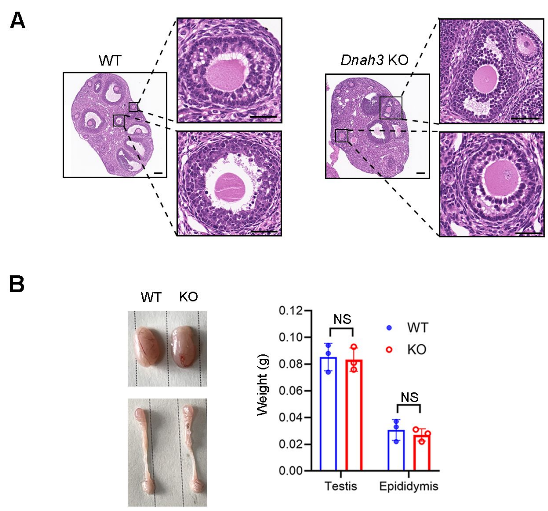


Figure S6

Fertility of *Dnah3* KO mice.

(**A**) H&E staining of ovary tissue sections from 8-week-old *Dnah3* KO female mice and WT female mice. Scale bars, 75 μm (n = three biologically independent WT mice or KO mice). (**B**) Sizes of the testis and epididymis of the 8-week-old *Dnah3* KO and WT mice (n = three biologically independent WT mice or KO mice; Student's *t* test; NS, no significance; error bars, s.e.m.).

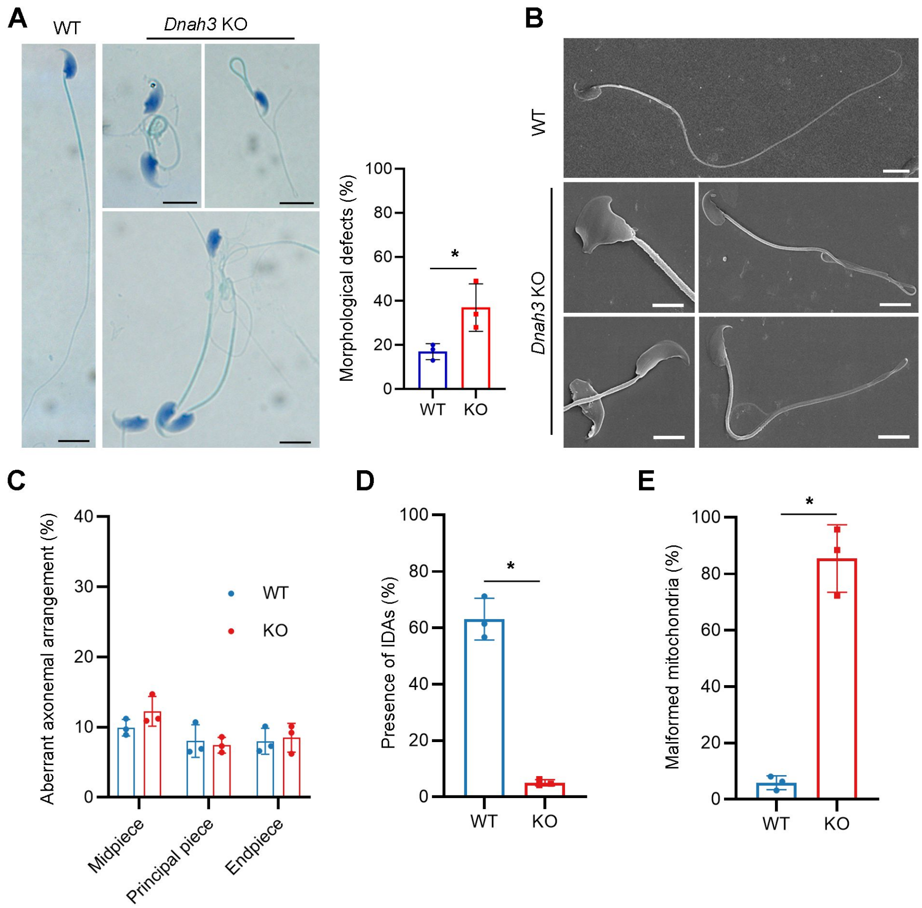


Figure S7

Morphology and ultrastructure of sperm isolated from *Dnah3* KO mice. (A, B)

Papanicolaou staining (A) and SEM analysis (B) revealed morphological defects in partial spermatozoa from *Dnah3* KO mice compared to WT mice. Scale bars in (A), 5 μm; scale bars in (B), 2.5 μm. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.). (**C**) The percentage of aberrant axonemal arrangement in different cross-sections of sperm from WT mice and *Dnah3* KO mice. (**D**) The percentage of microtubule doublets that presented IDAs in WT mice and *Dnah3* KO mice. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.). (**E**) Statistics of malformed mitochondria in the midpiece of sperm from WT mice and *Dnah3* KO mice. (n = three biologically independent WT mice or KO mice; Student's *t* test; error bars, s.e.m.).

Figure S8

Immunofluorescence staining of ODA-associated proteins in spermatozoa obtained from variants within *DNAH3* patients.

(**A** – **C**) The expression of DNAH8 (A), DNAH17 (B) and DNAI1 (C) in spermatozoa of the patients was comparable to that in normal controls. Red, DNAH8 in (A), DNAH17 in (B), DNAI1 in (C); green, α-Tubulin; blue, DAPI; scale bars, 5 μm.

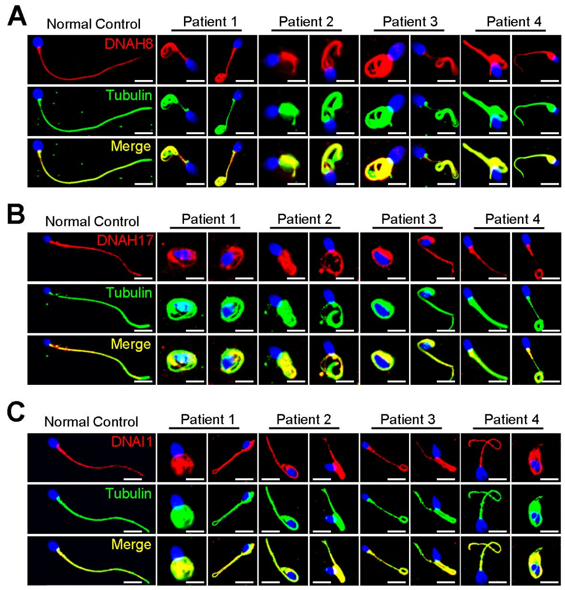
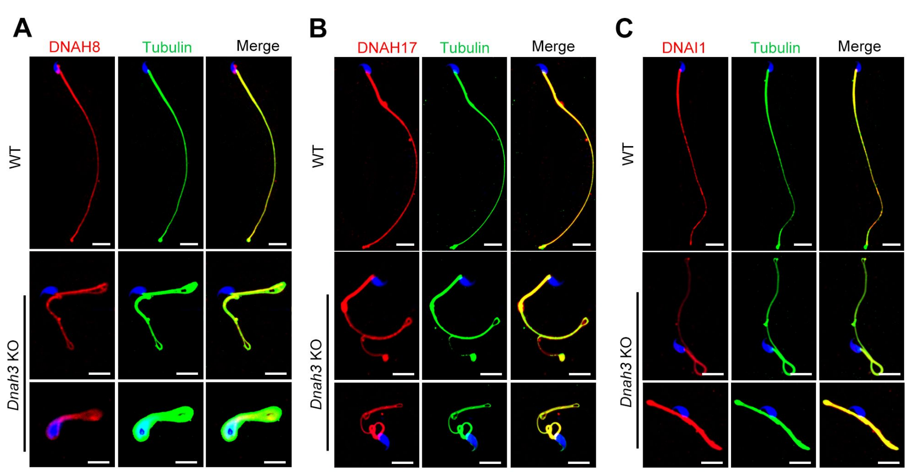


Figure S9

Immunofluorescence staining of ODA-associated proteins in spermatozoa of *Dnah3* KO and WT mice

(**A** – **C**) The expression of DNAH8 (A), DNAH17 (B) and DNAI1 (C) in spermatozoa from *Dnah3* KO mice was comparable to that in spermatozoa from WT mice. Red, DNAH8 in (A), DNAH17 in (B), DNAI1 in (C); green, α-Tubulin; blue, DAPI; scale bars, 5 μm.



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Editors

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Senior Editor
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The Lundquist Institute, Torrance, United States of America

Reviewer #1 (Public Review):

Summary:

Wang and colleagues identify biallelic variants of DNAH3 in four unrelated Han Chinese infertile men through whole-exome sequencing, which contributes to abnormal sperm flagellar morphology and ultrastructure. To investigate the importance of DNAH3 in male infertility, the authors generated crispant Dnah3 knockout (KO) male mice. They observed that KO mice are also infertile, showing a severe reduction in sperm movement with abnormal IDA (inner dynein arms) and mitochondrion structure. Moreover, nonfunctional DNAH3 expression decreased the expression of IDA-associated proteins in the spermatozoa of patients and KO mice, which are involved in the disruption of sperm motility. Interestingly, the infertility of patients and KO mice is rescued by intracytoplasmic sperm injection (ICSI). Taken together, the authors propose that DNAH3 is a novel pathogenic gene for asthenoterozoospermia and male infertility.

Strengths:

This work investigates the role of DNAH3 in sperm mobility and male infertility. By using gold-standard molecular biology techniques, the authors demonstrate with exquisite resolution the importance of DNAH3 in sperm morphology, showing strong evidence of its role in male infertility. Overall, this is a very interesting, well-written, and appealing article. All aspects of the study design and methods are well described and appropriate to address the main question of the manuscript. The conclusions drawn are consistent with the analyses conducted and supported by the data.

Weaknesses:

The paper is solid, and in its current form, I have not detected relevant weaknesses.

https://doi.org/10.7554/eLife.96755.2.sa3

Reviewer #2 (Public Review):

Wang et al. investigated the role of dynein axonemal heavy chain 3 (DNAH3) in male infertility. They found that variants of DNAH3 were present in four infertile men, and the deficiency of DNAH3 in sperm affects sperm mobility. Additionally, they showed that Dnah3 knockout male mice are infertile. Furthermore, they demonstrated that DNAH3 influences inner dynein arms by regulating several DNAH proteins. Importantly, they showed that intracytoplasmic sperm injection (ICSI) can rescue the infertility in Dnah3 knockout mice and two patients with DNAH3 variants.

Strengths:

The conclusions of this paper are well-supported by data.

Weaknesses:

The sample/patient size is small; however, the findings are consistent with those of a recent study on DNAH3 in male infertility involving 432 patients.

https://doi.org/10.7554/eLife.96755.2.sa2

Reviewer #3 (Public Review):

Summary:

(1) To further explore the genetic basis of asthenoteratozoospermia, the authors performed whole-exome sequencing analyses among infertile males affected by asthenoteratozoospermia. Four unrelated Han Chinese patients were found to carry biallelic variations of DNAH3, a gene encoding IDA-associated protein.
(2) To verify the function of IDA associated protein DNAH3, the authors generated a Dnah3-KO mouse model and revealed that the loss of DNAH3 leads to severe male infertility as a result of the severe reduction in sperm movement with the abnormal IDA and mitochondrion structures.
(3) Mechanically, they confirmed decreased expression of IDA-associated proteins (including DNAH1, DNAH6 and DNALI1) in the spermatozoa from patients with DNAH3 mutations and Dnah3-KO male mice.
(4) Then, they also found that male infertility caused by DNAH3 deficiency could be rescued by intracytoplasmic sperm injection (ICSI) treatment in humans and mice.

Strengths:

(1) In addition to existing research, the authors provided novel variants of DNAH3 as important factors leading to asthenoteratozoospermia. This further expands the spectrum of pathogenic variants in asthenoteratozoospermia.
(2) By mechanistic studies, they found that DNAH3 deficiency led to decreased expression of IDA-associated proteins, which may be used to explain the disruption of sperm motility and reduced fertility caused by DNAH3 deficiency.
(3) Then, successful ICSI outcomes were observed in patients with DNAH3 mutations and Dnah3 KO mice, which will provide an important reference for genetic counselling and clinical treatment of male infertility.

https://doi.org/10.7554/eLife.96755.2.sa1

Author response:

The following is the authors' response to the original reviews.

> *(1) Combined Public Reviews:*
>
> *Strengths:*
>
> *This work investigates the role of DNAH3 in sperm mobility and male infertility and utilised gold-standard molecular biology techniques, showing strong evidence of its role*

> *in male infertility. All aspects of the study design and methods are well described and appropriate to address the main question of the manuscript. The conclusions drawn are consistent with the analyses conducted and supported by the data.*

We extend our sincere gratitude to the expert reviewers for their valuable comments and insightful suggestions.

> *Weaknesses:*
>
> *(1.1) The manuscript lacks a comparison with previous studies on DNAH3 in the Discussion section.*

We thank the reviewers' comments.

Recently, Meng et al. identified bi-allelic variants in DNAH3 from patients diagnosed with asthenoteratozoospermia, revealing multiple morphological defects and a disrupted "9+2" arrangement in the patients' sperm (https://doi.org/10.1093/hropen/hoae003, PMID: 38312775). Furthermore, they generated *Dnah3* KO mice, which were infertile, and exhibited moderate morphological abnormalities with a normally structured “9 + 2” microtubule arrangement. In our study, we also observed similar phenotypic differences between the phenotypes of *DNAH3*-deficient patients and *Dnah3* KO mice. These findings indicate that DNAH3 may play crucial yet distinct roles in human and mouse male reproduction. Additionally, our TEM analysis demonstrated a notable absence of IDAs in sperm from both *DNAH3*-deficent patients and *Dnah3* KO mice, resembling the findings of Meng et al. To further investigate, we conducted immunofluorescent staining and western blotting to assess the levels of IDA-associated proteins (DNAH1, DNAH6 and DNALI1) and ODA-associated proteins (DNAH8, DNAH17 and DNAI1) in sperm samples from both our *DNAH3*-deficient patients and *Dnah3* KO mice. Our data revealed a reduction in IDA-associated protein levels and comparable ODA-associated protein levels in comparison to normal controls and WT mice, respectively, thus corroborating the TEM observations. These results suggest that DNAH3 is involved in sperm flagellar development in human and mice, specifically through its role in the assembly of IDAs.

Intriguingly, in our study, none of the patients with *DNAH3* deficiency reported experiencing any of the principal symptoms associated with PCD. Additionally, our *Dnah3* KO mice exhibited normal ciliary development in the lung, brain, eye, and oviduct. Similarly, Meng et al. did not mention any PCD symptoms in their *DNAH3*-deficient patients, and their *Dnah3* KO mice also demonstrated normal ciliary morphology in the trachea and brain. These combined observations suggest that DNAH3 may play a more significant role in sperm flagellar development than in other motile cilia functions. Given that DNAH3 is expressed in ciliary tissues, its role in these tissues remains intriguing and could be elucidated through sequencing of larger cohorts of individuals with PCD.

We have added these discussions in line 267 to 283, and line 300 to 303.

> *(1.2) The variants of DNAH3 in four infertile men were identified through whole-exome sequencing. Providing an overview of the WES data would be beneficial to offer additional insights into whether other variants may contribute the infertility. This could also help explain why ICSI only works for two out of four patients with DNAH3 variants.*

We thank the reviewer's helpful suggestions.

We have deposited the raw whole-exome sequencing data in the National Genomics Data Center (NGDC) (https://ngdc.cncb.ac.cn/, accession number: HRA007467). The clean reads, sequencing depth, sequencing coverage, and mapping quality of the WES on the patients are listed below (Table R1). A summary of WES has been presented in Table S1.

Author response table 1.

Quality of whole exome sequencing on infertile men.

Patients	Clean reads	Clean base	Average depth	Coverage (20X)	Mapping quality	Quality control
Patient 1	89.5 million	13.86 G	149	98.70%	99.93%	Success
Patient 2	34.6 million	9..99 G	63.00	98.48%	99.79%	Success
Patient 3	142.1 million	21.70 G	195	99.70%	99.97%	Success
Patient 4	73.8 million	11.32 G	102	98.90%	99.98%	Success

The variants identified through WES were annotated and filtered using Exomiser. Next, the variants were screened to obtain candidate variants based on the following criteria: (1) the allele frequency in the East Asian population was less than 1% in any database, including the ExAC Browser, gnomAD, and the 1000 Genomes Project; (2) the variants affected coding exons or canonical splice sites; (3) the variants were predicted to be possibly pathogenic or damaging.

Following filtering and screening, the numbers of candidate variants obtained were as follows: Patient 1: 98, Patient 2: 101, Patient 3: 67, and Patient 4: 91(Table S1). Subsequently, we utilized the Human Protein Atlas (HPA) database (https://www.proteinatlas.org/) and Mouse Genome Informatics (MGI) database (https://informatics.jax.org/) to analyze the expression patterns of corresponding genes. Variants whose corresponding genes were not expressed in the human or mouse testis were excluded from further consideration. We also consulted OMIM database and reviewed relevant literature to exclude variants associated with diseases unrelated to male infertility. Additionally, considering the assumption of a recessive inheritance pattern, we excluded all monoallelic variants. Ultimately, only bi-allelic variants in DNAH3 (NG_052617.1, NM_017539.2, NP_060009.1) remained, suggesting as the pathogenic variants responsible for the infertility of the patients (Table S1). These *DNAH3* variants were verified by Sanger sequencing on DNA from the patients' families.

We have added the overview of the WES in Table S1 and supplemented the analysis process of WES data in line 100 to 106, and line 348 to 360.

Additionally, we did not identify any pathogenic variants that associated with fertilization failure and early embryonic development in the two patients with failed ICSI outcomes. Therefore, these different ICSI outcomes might be attributed to additional unexplained factors from the female partners.

> *(1.3) Quantification of images would help substantiate the conclusions, particularly in Figures 2, 3, 4, and 6. Improved images in Figures 3A, 4B, and 4C, would help increase confidence in the claims made.*

In response to reviewer’s valuable suggestions. We presume that the reviewer means quantification of images in Figure S6, but not Figure 6.

We have compiled statistics for results shown in Figures 2, 3, 4, and S6. Specifically:

- The percentages of abnormal flagellar morphology in normal control and patients, associated with the observations in Figure 2A, have been shown in Figure S1A.

- The percentages of aberrant axonemal ultrastructure in different cross-sections of sperm from in normal control and patients, correspond to the findings in Figure 3A, have been presented in Figure S1B.

- The percentages of abnormal flagellar morphology in WT mice and *Dnah3* KO mice have been shown in Figure S7A.

- The percentages of aberrant axonemal arrangement in different cross-sections of sperm from WT mice and *Dnah3* KO mice, corresponding to the findings in Figure 4B, have been presented in Figure S7C.

- The percentages of microtubule doublets presenting IDAs in sperm from WT mice and *Dnah3* KO mice, related to Figure 4B, have been detailed in Figure S7D.

- The percentages of malformed mitochondria in the midpiece of sperm from WT mice and *Dnah3* KO mice, associated with the observations in Figure 4C, have been presented in Figure S7E.

Moreover, we have revised Figures 3A, 4B, and 4C by replacing the unclear TEM images.

> ***(2) Reviewer #1 (Recommendations for The Authors):***
>
> *(2.1) Please add reference(s) that support what is claimed in lines 83-84.*

We are very grateful for the reviewer's careful comments, we have added a reference that describing the homology and expression of DNAH3.

> *(2.2) In line 286, change "suggested" to "suggest".*

Thanks for the reviewer's comments. We have corrected the grammar.

> *(2.3) Please add reference(s) that support what is claimed in lines 359-360.*

According to the reviewer's suggestions, we have included references detailing the STA-PUT velocity sedimentation for isolation of single human and mouse testicular cells.

> *(2.4) In line 365, change "in" to "into".*

Thanks for the reviewer's careful comments, we have corrected this word.

> *(2.5) In Figure 7, I suggest changing "patients" to "wife or partners of patient". Given that the results are indeed from the spouses of the infertile men, I suggest making this small change to keep the consistency and clarity of what the authors did.*

In response to reviewer's kind suggestions, we have replaced "Patient" by "partners of Patient" and revised Figure 7.

> ***(3) Reviewer #2 (Recommendations for The Authors):***
>
> *(3.1) A summary of the WES data would be needed (i.e. number of reads, mapping quality, etc). As mentioned in the public review, it would be beneficial to present a summary of all variants identified in the data and clarify whether DNAH3 is the only gene that contains variants and whether these variants have been validated.*

Many thanks for reviewer's kind suggestions.

The clean reads, sequencing depth, sequencing coverage, and mapping quality of the WES on the patients are listed (see author response table 1) A summary of WES has been presented in Table S1.

The variants identified through WES were annotated and filtered using Exomiser. Next, the variants were screened to obtain candidate variants based on the following criteria: (1) the allele frequency in the East Asian population was less than 1% in any database, including the ExAC Browser, gnomAD, and the 1000 Genomes Project; (2) the variants affected coding exons or canonical splice sites; (3) the variants were predicted to be possibly pathogenic or damaging.

Following filtering and screening, the numbers of candidate variants obtained were as follows: Patient 1: 98, Patient 2: 101, Patient 3: 67, and Patient 4: 91(Table S1). Subsequently, we utilized the Human Protein Atlas (HPA) database (https://www.proteinatlas.org/) and Mouse Genome Informatics (MGI) database (https://informatics.jax.org/) to analyze the expression patterns of corresponding genes. Variants whose corresponding genes were not expressed in the human or mouse testis were excluded from further consideration. We also consulted OMIM database and reviewed relevant literature to exclude variants associated with diseases unrelated to male infertility. Additionally, considering the assumption of a recessive inheritance pattern, we excluded all monoallelic variants. Ultimately, only bi-allelic variants in DNAH3 (NG_052617.1, NM_017539.2, NP_060009.1) remained, suggesting as the pathogenic variants responsible for the infertility of the patients (Table S1). These *DNAH3* variants were verified by Sanger sequencing on DNA from the patients' families.

We have added the overview of the WES in Table S1 and supplemented the analysis process of WES data in line 100 to 106, and line 348 to 360.

> *(3.2) It would be beneficial to the scientific community if the raw data of WES could be uploaded to a public data repository, such as GEO.*

According to the reviewer's suggestion, we have deposited the raw whole-exome sequencing data in the National Genomics Data Center (NGDC) (https://ngdc.cncb.ac.cn/, accession number: HRA007467) and described its availability in the "Data Availability" section.

> *(3.3) In line 115, it is not clear how the prediction was made. Clarifying them by adding citations or describing methods that predict these pathways/functions would help strengthen it.*

Thanks for the reviewer's comments.

SIFT, PolyPhen-2, MutationTaster and CADD assess the deleteriousness of genetic variants by considering genomic features and evolutionary constraint of the surrounding sequence or structural and chemical property altercations by the amino acid substitutions. We have added websites and references of these tools in the manuscript (line 116 to 118).

Here are the principles of these tools.

- The SIFT considers the position at which the change occurred and the type of amino acid change, and then to predict whether an amino acid substitution in a protein will affect protein function [https://sift.bii.a-star.edu.sg/, PMID: 12824425].

- The PolyPhen-2 predicts the impact of an amino acid substitution on a human protein by considering several features, including sequence, phylogenetic, and structural information [http://genetics.bwh.harvard.edu/pph2/, PMID: 20354512].

- The MutationTaster utilizes a Bayes classifier to predict the functional consequences of amino acid substitutions, intronic and synonymous changes, short insertions/deletions (indels), etc. [https://www.mutationtaster.org/, PMID: 24681721].

- The CADD scores are based on diverse genomic features derived from surrounding sequence context, gene model annotations, evolutionary constraint, epigenetic measurements, and functional predictions [https://cadd.gs.washington.edu/, PMID: 30371827].

> ***(4) Reviewer #3 (Recommendations for The Authors):***
>
> *(4.1) Please ensure that all gene names used in your manuscript have been approved by the HUGO nomenclature committee. For example, "c.3590C>T (p.P1197L)" should be described as "c.3590C>T (Pro1197Leu)".*

In response to the reviewer's suggestion, we have improved all the names of gene and variants according to the HUGO nomenclature committee and HGVS Variant Nomenclature Committee, respectively.

> *(4.2) For Table 1, the authors should provide the rates of abnormal sperm morphologies using the sperm cells from normal male controls.*

Thanks for the reviewer's careful comments. Consistent with the WHO laboratory manual (*World Health Organization. WHO laboratory manual for the examination and processing of human semen. World Health Organization, 2021.*), our routine semen analysis establishes 4% as the minimum rate of sperm with normal morphology but does not define the maximum rate of various tail defects. However, we reviewed the routine semen analysis on the normal controls in our study, and the approximate distribution of sperm with various flagellar in the normal controls was as follows: normal flagella, 78.6%; absent flagella, 1.7%; short flagella, 0.6%; coiled flagella, 12.5%; bent flagella, 7.9%; irregular flagella, 1.8%.

> *(4.3) In Table 2, "Mutation Tester" or "Mutation Taster"?*

We thank the reviewer's comments. It should be "MutationTaster", and we have corrected this mistake in Table 2 and the manuscript.

> *(4.4) In Figure 2B, the bars for patient 1 should be aligned.*

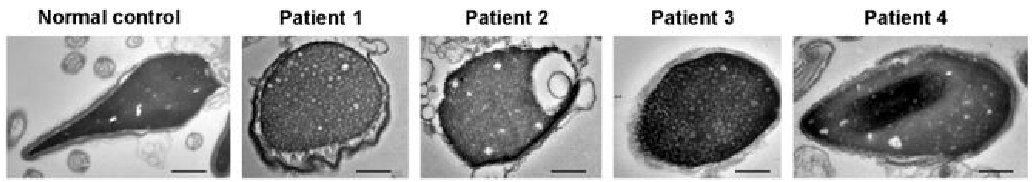
Following the reviewer's valuable suggestion, we have ensured consistent scar bar alignment in Figure 2B and implemented this alignment throughout all other figures.

> *(4.5) In Figure 3A, what about the ultrastructure for sperm heads in DNAH3 deficient sperm cell? The authors previously mentioned abnormalities in sperm head morphologies (Figure 2B) in patients with DNAH3 mutations.*

We thank the reviewers for their kind comments. A small fraction of abnormal sperm head of our patients was captured under TEM, manifested by round head with loose chromatin (Author response image 1)

Author response image 1.

Ultrastructure of sperm head from DNAH3-deficient infertile men. TEM analysis revealed a fraction of round head with loose chromatin in patients harboring *DNAH3* variants. Scale bars, 200 nm.



> *(4.6) In Figure S6, the authors should provide the rates of abnormal sperm morphologies for* Dnah3 *KO male mice.*

In response to the reviewer's valuable suggestion, we have quantified morphological defects in spermatozoa from both *Dnah3* KO and WT mice. Compared to about 17% morphological abnormalities in sperm from WT mice, the morphological abnormalities in sperm from *Dnah3* KO mice were about 37%. The results are presented in the revised Figure S7.

https://doi.org/10.7554/eLife.96755.2.sa0